

UNIVERSITY OF CALIFORNIA SAN DIEGO

Increasing the Signal Efficiency of $H \rightarrow \gamma\gamma$ and $HH \rightarrow b\bar{b}\gamma\gamma$ Analyses Using Machine Learning
Techniques

A dissertation submitted in partial satisfaction of the
requirements for the degree Doctor of Philosophy

in

Physics

by

Robert John Lee

Committee in charge:

Professor James G. Branson, Chair
Professor Javier Duarte
Professor Michael Gilson
Doctor Marco Pieri
Professor Yi-Zhuang You

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University of California San Diego

2026

DEDICATION

Probably no dedication.

EPIGRAPH

True ease in writing comes from art, not chance,
As those move easiest who have learn'd to dance.
'T is not enough to no harshness gives offence,—
The sound must seem an echo to the sense.

Alexander Pope

You write with ease to show your breeding,
But easy writing's curst hard reading.

Richard Brinsley Sheridan

Writing, at its best, is a lonely life. Organizations for writers palliate the writer's loneliness, but I doubt if they improve his writing. He grows in public stature as he sheds his loneliness and often his work deteriorates. For he does his work alone and if he is a good enough writer he must face eternity, or the lack of it, each day.

Ernest Hemingway

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PREFACE

Probably no preface

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Acknowledgements

VITA

- 2015–2018 Teaching Assistant, Department of Physics
Arizona State University
- 2015–2018 Research Assistant
Arizona State University Meson Physics Group
- 2018 Bachelor of Science
Arizona State University Barret Honor's College
- 2018–2019 Teaching Assistant, Department of Physics
University of California, San Diego
- 2019–2026 Graduate Student Researcher, University of California, San Diego
- 2026 Doctor of Philosophy, University of California, San Diego

ABSTRACT OF THE DISSERTATION

Increasing the Signal Efficiency of $H \rightarrow \gamma\gamma$ and $HH \rightarrow b\bar{b}\gamma\gamma$ Analyses Using Machine Learning Techniques

by

Robert John Lee

Doctor of Philosophy in Physics

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Professor James G. Branson, Chair

Abstract

Introduction

First theorized in 1964, the Higgs field constitutes a vital component of the Standard Model (SM) of particle physics. The insertion of this field to the SM allowed a mechanism for the spontaneous breaking of the electroweak symmetry, a necessary fix to the versions of the antecedent theories of fundamental particles. This spontaneous symmetry breaking provides an avenue through which the fundamental particles may gain their masses; until the formulation of the Higgs fields, the prevailing version of the SM could not account for the existence of – let alone the values of – the masses of quarks and the force-carrying gauge bosons. Luckily, the prediction of the Higgs field proposed a corresponding bosonic particle to mediate the interactions of this field. As the fundamental particles predicted by the SM were discovered throughout the 20th century, the Higgs boson proved to be elusive, in fact eventually becoming the last undiscovered fundamental particle predicted by the SM.

Since the discovery of the Higgs boson in 2012, the Large Hadron Collider has undertaken continuously more precise measurements of its properties. As the amount of available data has accumulated, increasingly rare channels of analysis have become explorable. To enable these high-quality analyses, the hardware and software of the LHC and its detectors has had to be improved considerably. Due to the abundance of background processes created at the high energies achieved by the LHC, coupled with the relative rarity of the processes by which the Higgs boson is produced and decays, complex software is necessary to disentangle the signal processes.

This software is constructed using a layered approach, where each subsequent step in the processing of the data serves to further reduce the total amount of data events being analyzed,

while, ideally, preserving as many events considered signal as possible. The first stages of this data reduction are achieved by a set of triggers. These triggers are relatively coarse-grain, seeking to reject the most obvious background events and thereby reduce the amount of data being stored and analyzed by three to four orders of magnitude. Of course, given the large number of different final states of the Higgs boson decay, many different triggers must be developed to dig out the specific signal signature required for a specific analysis. Once these triggers are applied, further analysis software is used on the resulting data to produce final results of the measurements of the various properties of the desired physics process.

The goal of this paper is to examine the improvement of a specific set of these triggers and analysis-level cuts. While many channels of Higgs boson decay exist, the decay into two photons has been proven to be a major component of the analysis of the Higgs boson; this process provides a relatively clean decay signature, and occurs at a high enough rate to provide ample statistics for analysis.

Chapter 1

The Standard Model and Higgs Boson

Theory Citations: [7]: Short 1999 Overview of SM. [11]: 2024 review of particle physics. [2]: 2012 Higgs Discovery phys review letters publication. [3] 2013 JHEP Higgs discovery publication. [1] 2022 CMS Higgs 10 year review.

Detector Citations: [4] 2008 Early CMS Description. [8] 2024 Run3 CMS Description (use this mainly). [5] 2008 Overall LHC design.

Trigger Citations: [13] 2020 L1 Trigger review. [10] 2017 overall trigger structure. [9] 2024 review of Run2 HLT performance.

The standard model (SM) of particle physics is the quantum gauge field theory that describes all currently known physical interactions besides gravity. Early formulations considered there to be three distinct gauge theories that ruled interactions between physical objects: Quantum Electrodynamics (QED), which describes the electromagnetic force between charged objects; Quantum Chromodynamics (QCD), which describes the strong nuclear force; and Weak Theory, which describes the weak nuclear force.

Within the SM, there are two major forms of particle. Spin $1/2$ particles, called fermions, comprise all matter as we know it, and include quarks and leptons. Integer spin particles, called bosons, act as the force carriers that mediate interactions between these fermions. QED posits the photon as the gauge boson, which mediates interactions between all electrically charged particles – including both charged leptons and quarks. QCD describes eight gluons as the force

mediators between color-charged quarks. The weak theory describes the massive $W^\pm$ and Z^0 bosons as the mediators of the weak force, which is responsible for radioactive decay.

As will be discussed in 1.3, experimental findings pointed early theorists towards the unification of the electromagnetic and weak forces into what is now known as the electroweak (EW) theory. The

1.1 Quantum Electrodynamics (QED)

Use Gaillard, Grannis, and Sciulli

1.2 Quantum Chromodynamics (QCD)

Use Gaillard, Grannis, and Sciulli AND PDG Physics summary page 153

1.3 Electroweak (EW) Theory

Use Gaillard, Grannis, and Sciulli AND PDG Physics summary page 185

1.4 The Higgs Field and Higgs Boson

Use PDG Physics summary page 211-218

1.4.1 Motivation for the Higgs Field

1.4.2 Properties of the Higgs Potential

1.4.3 Production Modes of a Single Higgs Boson

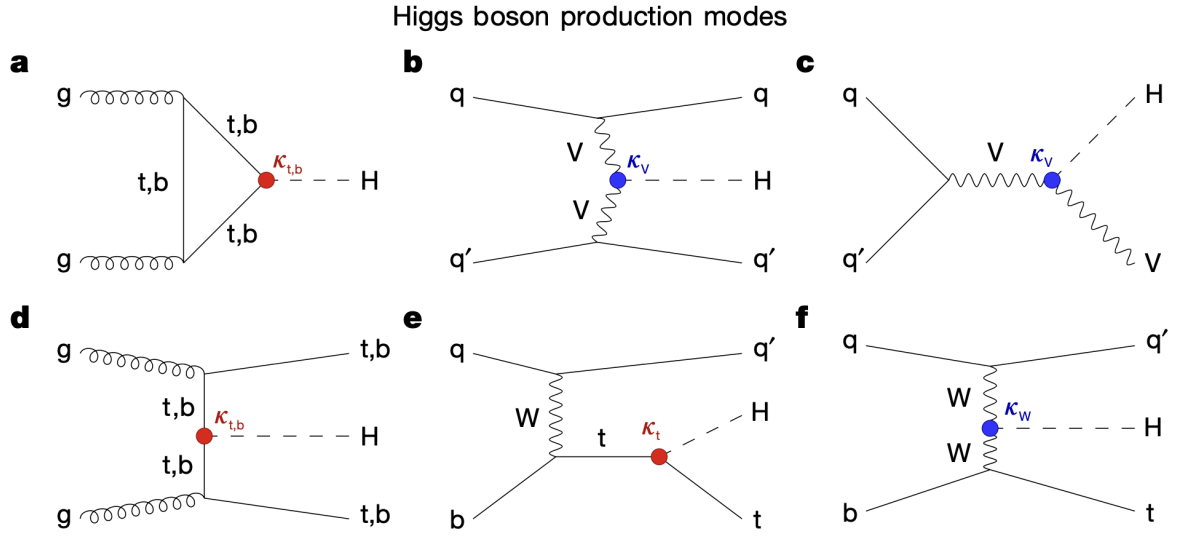


Figure 1.1. Feynman diagrams of known Higgs production modes. Reproduced from [CMS10YearReview]

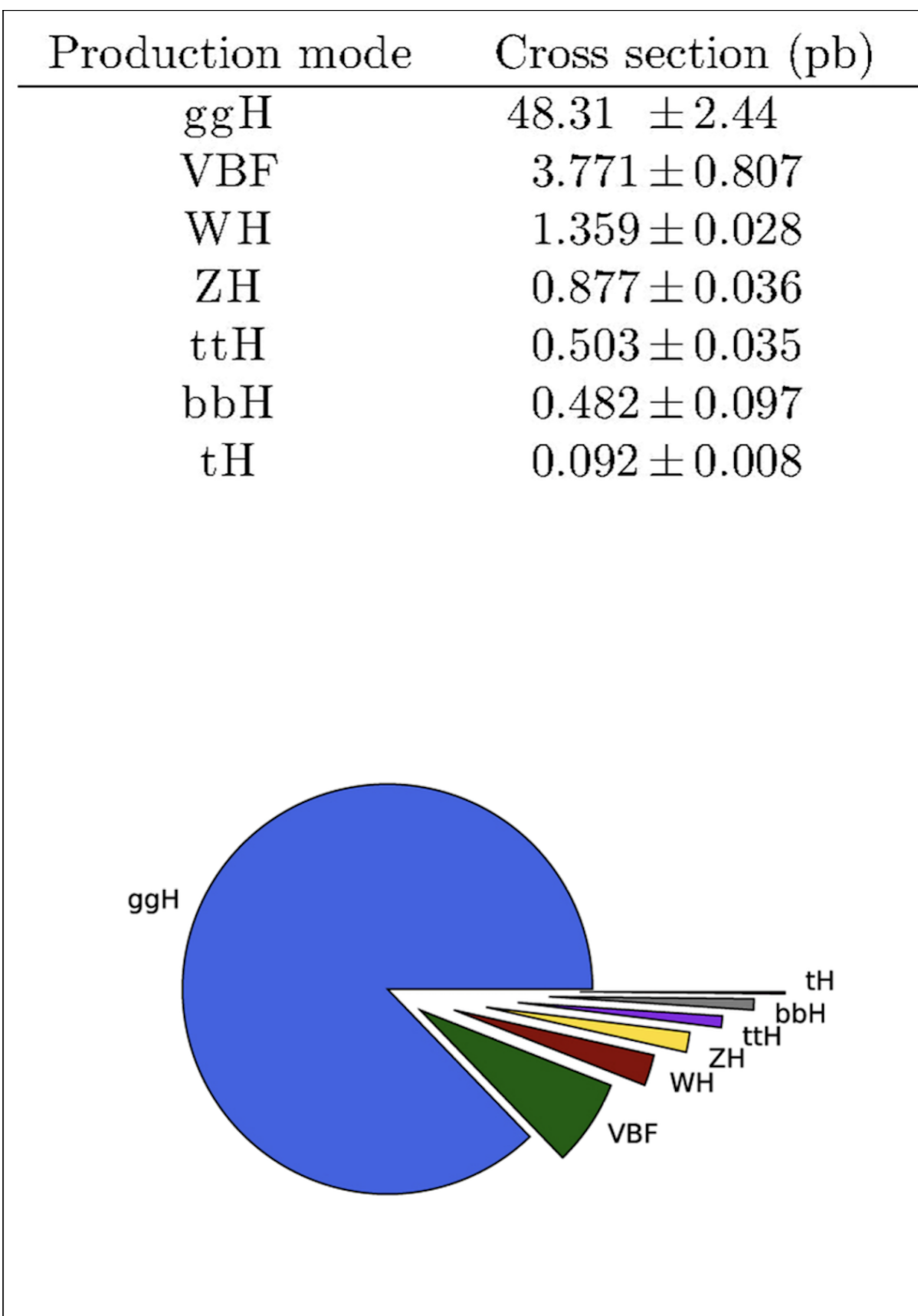


Figure 1.2. Production cross sections for Higgs creation. Reproduced from [CMS10YearReview]

1.4.4 Decay Modes of a Single Higgs Boson

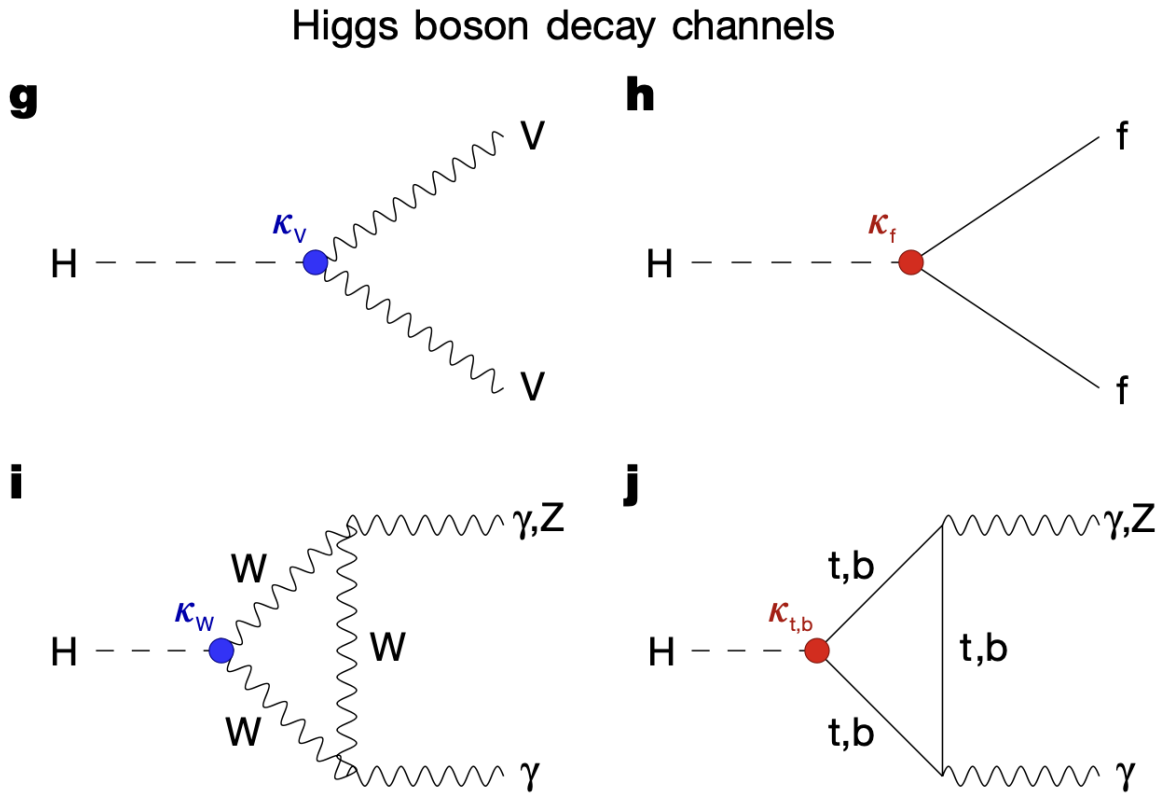


Figure 1.3. Feynman diagrams of known Higgs decay modes. Reproduced from [CMS10YearReview]

Decay channel	Branching fraction (%)	
bb	57.63	± 0.70
WW	22.00	± 0.33
gg	8.15	± 0.42
$\tau\tau$	6.21	± 0.09
cc	2.86	± 0.09
ZZ	2.71	± 0.04
$\gamma\gamma$	0.227	± 0.005
$Z\gamma$	0.157	± 0.009
ss	0.025	± 0.001
$\mu\mu$	0.0216	± 0.0004

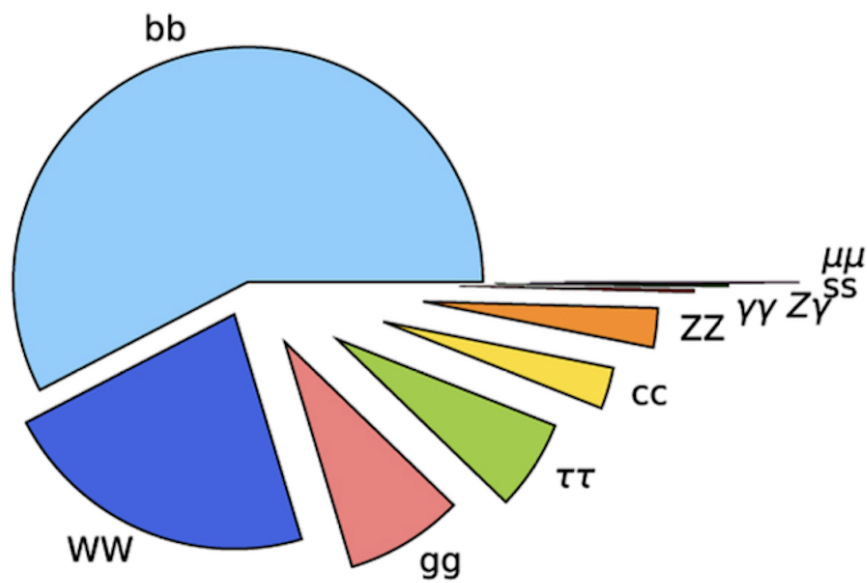


Figure 1.4. Decay branching fractions⁸ for a single Higgs. Reproduced from [CMS10YearReview]

1.5 Higgs Boson Decaying to Two Photons ($H \rightarrow \gamma\gamma$)

1.5.1 Useful Properties for Analysis

1.5.2 Historical Results

1.6 Di-Higgs Production and Properties

1.6.1 Di-Higgs Production Modes

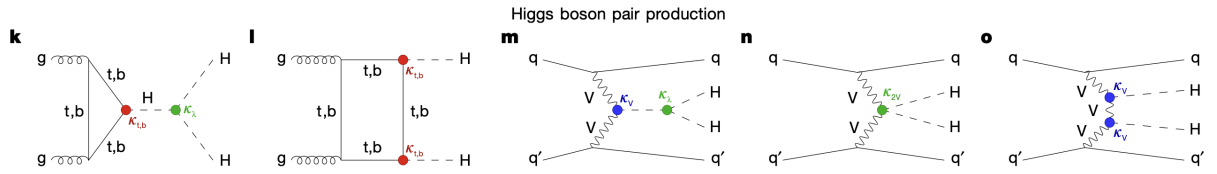


Figure 1.5. Feynman diagrams of known Higgs decay modes. Reproduced from [CMS10YearReview]

1.6.2 Di-Higgs Decay Modes

1.6.3 Usefulness of the $HH \rightarrow b\bar{b}\gamma\gamma$ Channel

Chapter 2

The LHC and CMS Detector

2.1 The Large Hadron Collider

The Large Hadron Collider, which replaced the preexisting Large Electron-Positron Collider, was constructed between 2000 and 2008. Designed to reach new energy frontiers by colliding protons at up to 14 TeV, the LHC provided particles physicists with their most powerful tool yet to test the Standard Model and beyond. At the most basic level, these proton-proton collisions are achieved by crossing two counter-rotating proton beams at specific points. These counter-rotating beams are accelerated along the 26.7 km circumference of the LHC within a tunnel of 3.7 m internal diameter. Initial designs proposed two completely separated proton beam pipes, but it became clear that the space requirements of the existing CERN tunnel (constructed for LEP) precluded this possibility. As a result, a "two-in-one" superconducting magnet design was adopted that allowed the two beams to be transported in separate magnetic coils sharing the same mechanical and cryostat space.

2.1.1 LHC Design

2.1.2 Collisions at the LHC

2.1.3 LHC Experiments

Collisions at the LHC occur at 4 major interaction points, where 4 separate detectors have been constructed to allow independent measurements of target physics processes. Two of these

detectors, ATLAS and the Compact Muon Solenoid (CMS), are designed as general-purpose detectors with systems allowing the measurement of a broad range of physics processes. Since these detectors are constructed using different technologies and detector elements, the LHC is able to produce separate physics results that can be cross-checked. A major motivation for the construction of these experiments was the detection of the Higgs boson in its various decay modes.

2.2 The Compact Muon Solenoid Detector

The Compact Muon Solenoid is designed as a 22 m cylinder with a total diameter of 15 m. Centered on the beampipe where a proton-proton bunch crossing occurs, the detector consists of various layers of detector elements designed to track and measure different resultant particles. The general structure of the detector will be described beginning at the center and moving outward. A schematic of the detector can be seen in Figure 2.1. Later sections will provide additional detail on elements of particular interest for our $H \rightarrow \gamma\gamma$ analyses.

After particles collide and are scattered, they first interact with the central beampipe that houses the incoming proton bunches. While this element does not provide any detecting capabilities, interactions between the outgoing particles and the material of the beampipe may cause additional scattering and particle production. Next, the particles enter the robust inner tracking system, designed to be able to reconstruct the path of charged particles through the detector. This inner tracking system is composed of an internal pixel detector surrounded by an external strip detector. These elements will be described in section 2.2.1.

The next detector element encountered is the Electromagnetic Calorimeter (ECal), which aims to measure the energies and locations of electrons and photons. Of course, given that the following analysis will be centered on photons, this detector element will be of great interest. This ECal is composed of scintillating $PbWO_4$ crystals arranged into both a cylindrical ECal Barrel (EB) concentric with the beamline and two circular ECal Endcaps (EE) oriented orthogonally to

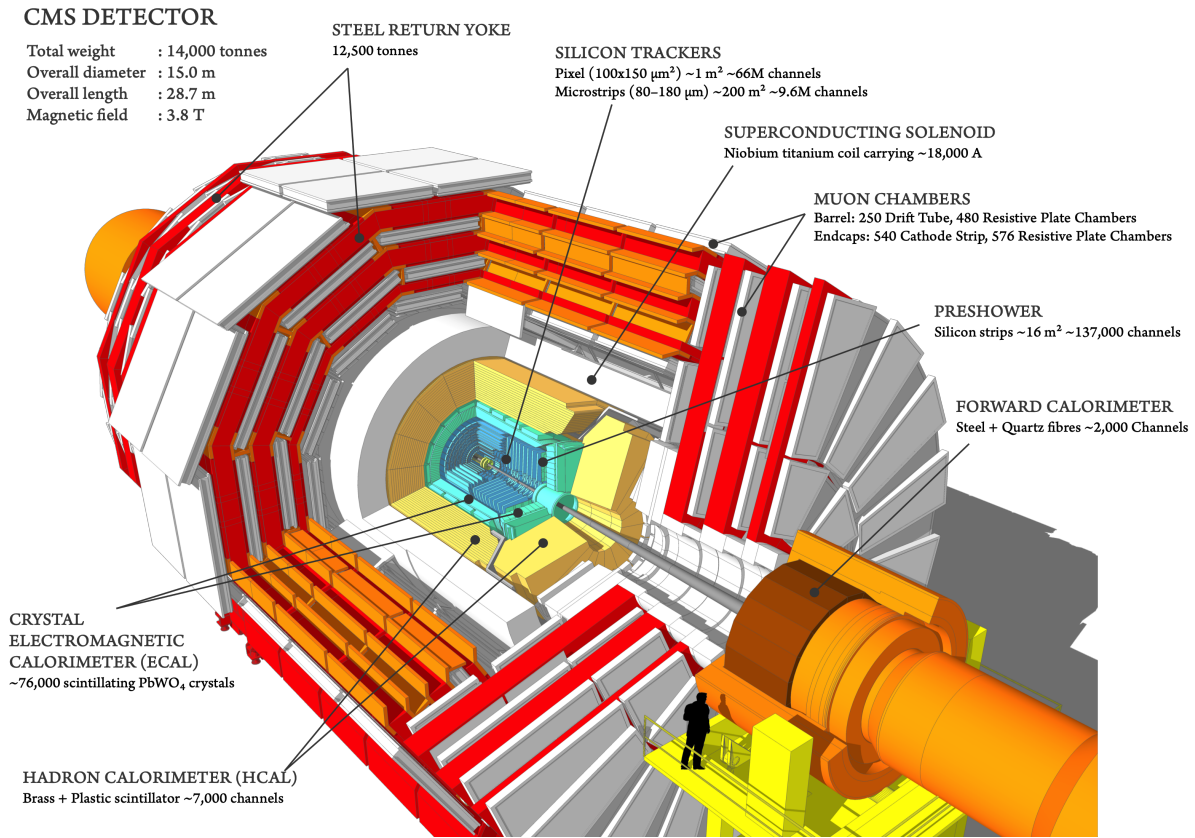


Figure 2.1. A full view of the CMS Detector, with each major system labeled. Reproduced from [CMSRun3Paper]

the beamline. A Preshower detector is inserted in front of the EE detectors. More details on this detector element will be provided in section 2.2.2.

Further from the beamline, the Hadronic calorimeter (HCal) aims to measure the locations and energies of charged and neutral hadronic particles using plastic scintillator tiles imbedded in a brass volume. Similarly to the ECal, the HCal is composed of a barrel (HB) region and two endcaps (EB). This detector will be described in more detail in section 2.2.3.

A superconducting solenoid encases all three of these internal regions, providing an internal magnetic field of 3.8T. This magnetic field allows the paths of charged particles to be bent, providing more precise measurement of their properties. While vital to the physics of the detector, this solenoid does not contain any detecting elements. Beyond this magnet lies the highly complicated Muon chambers, which will not be described for this analysis. Close to the

beamline but at the ends of the z-axis of the detector lies the forward calorimeter, which will also not be described for this analysis. The entire detector is shrouded in a steel return yoke.

2.2.1 Beampipe and Inner Tracking System

A major goal of the CMS detector is to reconstruct the tracks of charged particles. This allows the software to not only track and measure the flight of an individual particle, but it allows the primary interaction vertex (PV) to be measured more accurately. This is achieved by a two-layered approach; the pixel detectors closest to the beamline provide high-accuracy three-dimensional spatial information on the particles immediately after they leave the beampipe, while the strip detectors are used to reconstruct where the particle went after this precise initial location. Since the analysis of $H \rightarrow \gamma\gamma$ is focused on photons, charged tracks are not expected for the final state particles of interest. This does not mean they are of no use to us, however – charged tracks can be used to exclude particles from our final state analysis, allowing us to reduce the number of resultant particles and increase the likelihood that the final particles analyzed are photons.

Pixel Detector

The pixel detectors achieve high spatial resolution by using 124 million channels each composed of a single pixel. As with most detector elements at CMS, the pixel detectors are formed by a barrel region (BPIX) and two circular disk endcap regions (FPIX). The barrels use 4 layers of silicon pixels, located at 29, 68, 109, and 160 mm from the beamline. The endcap detectors use 3 layers of pixels, at distances of 291, 396, and 516 mm from the center of the detector along the z-axis. This layout ensures that any charged particle will strike the pixel trackers 4 times at varying distances from the interaction vertex, allowing precise reconstruction of the track starting at the edge of the beampipe. [?Table of number of modules for each pixel detector? Table 1 Run3CMSDetector]

The individual pixels, of the same size in both the BPIX and FPIX, are housed on sensor

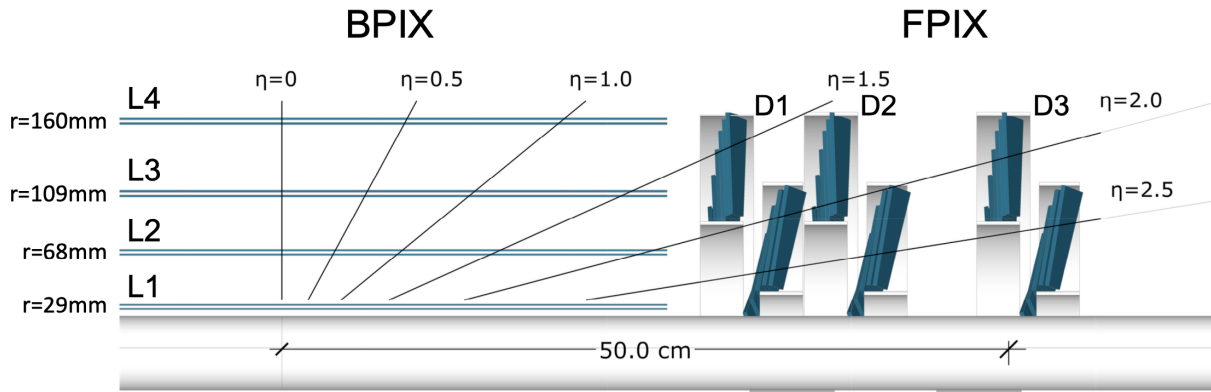


Figure 2.2. Longitudinal view of the pixel detector, where the beampipe is represented at the bottom of the image. Reproduced from [CMSRun3Paper]

modules with 160×416 pixels each (with a total size of $100 \times 150 \mu m^2$). This modular structure allows easy replacement of faulty modules, as well as allowing more efficient production. The BPIX contains 1184 such modules, across its 4 layers, whereas the FPIX contains 672 modules. Light-weight mechanical structures provide the mounting points for these modules, and include cooling apparatus.

Strip Detector

The innermost pixel detector is rather small, only extending 160mm radially from the beamline, and 516mm along the beam direction from the interaction point. The majority of the tracking system is composed of the silicon strip tracker (SST). The SST contains 9.3 million silicon micro-strips grouped into 15,148 modules. While there are far fewer channels in the SST, the volume covered is much larger; once the origin of a particle track is pinpointed by the pixel detectors, the strip detectors are able to detect the particle as it travels through a large volume of the detector. The SST has a diameter of 2.5m and a length along the beamline of 5m, so it is clearly a more coarse-grain tracking sensor.

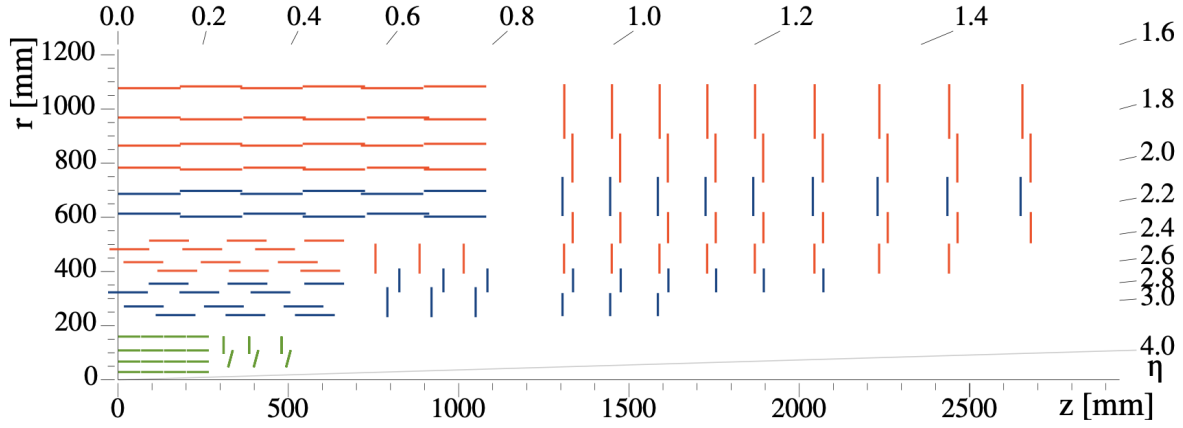


Figure 2.3. Full view of the tracker system. The inner pixel tracker is shown in green. Reproduced from [CMSRun3Paper]

2.2.2 Electromagnetic Calorimeter

Beyond the inner tracking system, the ECAL system is designed to measure the energies, locations, and arrival times of electrons and photons. Given that this paper will be dealing with a two-photon final state, the details of this detector are highly relevant. Similarly to the inner tracker, the ECAL is composed of modules of individual detector elements. Here, however, since the goal is to measure photons and electrons, scintillating material is required. As a result, the individual detector elements are composed of scintillating lead tungstate ($PbWO_4$) crystals. These crystals were chosen because, in addition to being optically clear, they are radiation-hard. Given that the detector will be exposed to high levels of radiation for long continuous stretches, it is imperative that the crystals need not be replaced frequently. The photons from the scintillation of these crystals are detected by photodetectors. The photodetectors differ, however, between the barrel and endcap as explained in detail below.

ECal Barrel (EB)

The barrel component of the ECAL (EB) is designed, of course, with cylindrical symmetry. In the azimuthal direction, ϕ , there are 360 crystals per "ring". The barrel detector covers

the pseudorapidity range of $|\eta| < 1.479$, and there are 180 crystals per "row" in this range. As a result, the EB consists of 61,200 total $PbWO_4$ crystals. These crystals are tapered, such that the scintillated photons are funneled to the back of the crystal where two avalanche photodiodes (APDs) per crystal collect them. The cross-sectional area of the crystal faces is $22 \times 22 \text{ mm}^2$. The face of each crystal lies at a distance of 1.29m from the center of the beamline.

As with the tracker, the individual crystals are combined into submodules and modules, where readout and power-supplying electronics are consolidated. These submodules are of an "aveolar" structure, where the dividing walls between individual crystals are 0.1mm thick and are made of a combination of aluminum and fibre-epoxy resin. This leads to a total crystal-to-crystal distance of 0.35mm within each submodule. This provides sufficient granularity while still ensuring complete geometric coverage with a reasonable number of individual channels.

ECal Endcaps (EE)

2.2.3 Hadron Calorimeter

2.2.4 Forward Calorimeter

2.2.5 The Muon Detector

Chapter 3

Triggering and Offline Software

In Run3 of the LHC, the increased luminosity leads to an average of 60 collisions per bunch crossing (termed "Pileup") (REF?). Given that these bunch crossings occur at a rate of 40MHz (every 25ns) and can create a large number of individual particles, the processing and storage facilities are incapable of saving and analyzing every interaction generated in each bunch crossing. The triggering system, therefore, is designed to drastically reduce the number of output events while preserving as many events matching signal signatures as possible. Overall, the triggers aim to reduce the data rate from 40MHz down to just 5kHz. This reduction is achieved by a two-tiered trigger system consisting of the Level 1 (L1) triggers and the High Level Triggers (HLT). Since there are many possible signal signatures depending on the analysis being conducted, there are many individual triggers at both L1 and HLT. Details of the L1 triggers will be given below in section 3.1, and the HLT will be described in great detail in section 3.3.

3.1 Level-1 Triggers

The Level 1 trigger is described in detail in [13], but the primary points are summarized in this section. As described in section 2.1.2, collisions in CMS during Run 3 occur at a rate of 40 million per second (40MHz). The goal of the L1 trigger is to select at most 100,000 of these events per second (100kHz) to pass on to the next level of the trigger, the HLT. The L1 trigger is designed to operate as quickly as possible, with a target latency of just 4 μ seconds. In order to

achieve this very low latency, minimal event reconstruction is done at L1; instead, this trigger performs minimal processing on the so-called trigger primitives (TP), which are the raw outputs from detector elements. The details of these TPs will be discussed below in section 3.1.1.

In order to process events as quickly as possible, software is eschewed in favor of an FPGA-based hardware system [10]. There are two of these systems corresponding to the two major detector systems: the L1 calorimeter trigger processes the information from both the Electron and Hadron Calorimeters, while the L1 muon trigger processes the information from the muon chambers. The calorimeter trigger system is of the greatest interest to this paper, as the photon candidates from $H \rightarrow \gamma\gamma$ events are largely measured by the ECal with some additional information also being drawn from the HCal. In section 3.1.2, then, the calorimeter trigger system will be discussed in detail.

These two systems consist of many individual triggers, known as the L1 menu. This menu is designed to target specific physics objects, and consists of several triggers for each object. Muon triggers comprise nearly half of this menu. Additional triggers exist for both τ leptons and the identification of jet objects. Finally there are triggers designed to measure electrons and photons (termed e/γ triggers). These triggers are, of course, the most important for $H \rightarrow \gamma\gamma$ processes, and the identification of photon candidates is the first step in finding and measuring these signal processes. The details of the L1 menu, with particular emphasis on the e/γ triggers, are presented below in section 3.1.4.

3.1.1 Information Available to L1 Triggers

The inputs to the L1 trigger are entirely in the form of trigger primitives. For the calorimeter trigger, these TPs are in the form of energy deposition in a specific detector element [13]. The HCal and ECal, detailed in section 2.2.3 and 2.2.2, are only able to measure the energy deposited in their individual elements. So-called Trigger Towers (TTs) group together the ECal and HCal elements in a specific region of a detector. For the barrel detector, a 5x5 set of EB crystals is combined with the corresponding HB tower behind them into a single TT. This gives

a barrel resolution of $\Delta\eta \times \Delta\phi = 0.087 \times 0.087$ radians. In the endcap, the presence of the HF detector complicates the TTs; the information from the EE, HE, and HF detectors must all be merged in a single TT per channel. Additionally, the geometry is complicated by the variable crystal size and spacing in the endcap. While the overall TT resolution consequentially varies in the endcap region, the maximum TT size is $\Delta\eta \times \Delta\phi = 0.17 \times 0.17$ radians.

The L1 TPs for the muon triggers are considerably more complex, as there are more detector elements to consider. The cathode strip chambers, drift tubes, and resistive plate chambers described in section 2.2.5 all overlap and must be combined in each TT. Furthermore, the TPs in the muon L1 system are not merely energy depositions. Instead, the muon detector subsystems provide energy, coordinate, and quality information that is later used to reconstruct a track.

3.1.2 The L1 Trigger Architecture

The calorimeter L1 trigger consists of two distinct layers of processing. Layer-1 receives the inputs directly from the TPs listed above, and sorts and calibrates them rapidly. Layer-2 takes the outputs from Layer-1 and performs primitive calibration and reconstruction, which allows the trigger to roughly identify physics objects such as electrons, τ leptons, and jets. A rather complex time-multiplexed hardware system is used in Layer-2, which allows the low, fixed latency necessary at the L1 trigger. Layer-2 consists of several separate reconstruction algorithms, of which primarily the e/γ reconstruction algorithm is of interest for the $H \rightarrow \gamma\gamma$ trigger. Details of this reconstruction will be provided in the next section. Once the objects are reconstructed, they are passed to the next stage in the trigger system.

The muon L1 trigger has 3 separate muon track finders (MTFs) which aim to reconstruct tracks in three distinct regions of the detector (barrel, endcap, and overlap). In the barrel, only the DT and RPC TPs are used to reconstruct tracks. In the endcap, only the CSC and RPC TPs are used to achieve the same goal. In the overlap region, all three subdetectors are combined. These MTFs feed their outputs directly to the global muon trigger (μ GMT), which receives up

to 108 muon tracks per event. The μ GMT then sorts the tracks and removes duplicates. The final output of the muon triggers has at most 8 muon tracks.

Both the Layer-2 calorimeter trigger and the μ GMT output to the Global Trigger (μ GT). The μ GT combines this information using a set of complex algorithms called the trigger menu described below in section 3.1.4. Six independent boards are used to distribute the processing of the L1 menu, and a final decision is rendered on whether to keep or discard an event. Below, in figure 3.1, a schematic of the full process from TPs to μ GT is reproduced from [13].

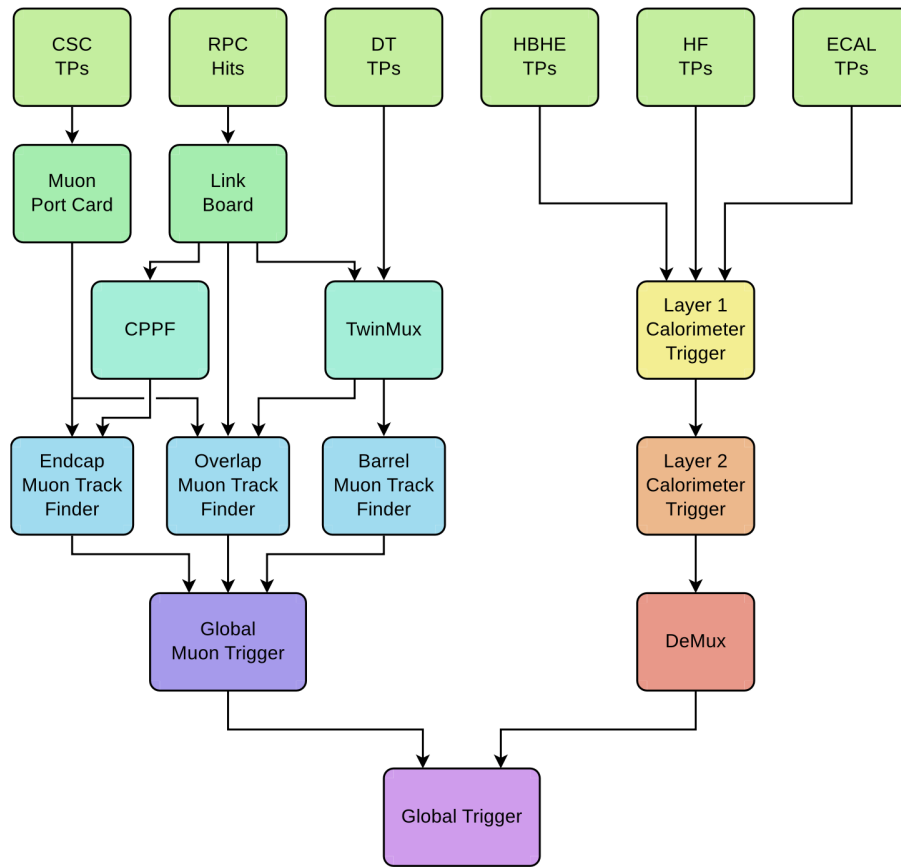


Figure 3.1. A diagram of the L1 Trigger architecture, where the lefthand tree corresponds to the muon triggers and the righthand tree to the calorimeter triggers. Reproduced from [13].

3.1.3 The L1 e/γ Reconstruction

Since photons and electrons have no associated tracking information, the only way that they can be identified is by their energy signatures.

3.1.4 The L1 Menu

3.2 The Data Acquisition System (DAQ)

3.3 High Level Trigger

The final step of triggering before data is output for analysis, the HLT aims to reduce the rate from the 100kHz output by the L1 trigger down to a total of just 5kHz [9]. In contrast to the hardware-based L1 trigger, the HLT is a software-based suite of triggers that runs on a dedicated server of processors called the HLT filter farm.

3.3.1 HLT Event Reconstruction

3.3.2 HLT Trigger Rates and Timing

3.3.3 HLT Software and ConfDB

3.4 Analysis Software

3.4.1 Offline Reconstruction

3.4.2 The NanoAOD File Format

3.4.3 NanoAOD Event Reconstruction

3.4.4 HiggsDNA

Chapter 4

Boosted Decision Trees and XGBoost

The newly-proposed trigger and complementary offline selection – discussed in chapters 5 and 6, respectively – is based on a machine learning construct called a Boosted Decision Tree (BDT). This ML method provides a robust and relatively simple algorithm for binary classification problems, which works well both with the HLT and with subsequent offline classification and event selection. The theoretical details and structure of BDTs is described in section 4.1. The software package XGBoost (short for eXtreme Gradient Boosting) will be used for both the creation and evaluation of the BDTs employed at the trigger and offline analysis levels. This software is described in more detail in section 4.3.

4.1 Gradient Descent Boosting

The original description of boosted decision trees and gradient descent boosting is described in "Greedy Function Approximation: A Gradient Boosting Machine" by Jerome H. Friedman [6]. The following section is largely based on this seminal paper, with emphasis placed on the facets of the technique most relevant to this paper.

At its core, a binary classification problem is a function estimation problem; a function must be constructed that maps some input values onto a single output score. The general parlance is to label the input variables for a single event as $\vec{x} = \{x_1, x_2, \dots, x_n\}$, where n is the total number of selected input values. The output, however, is a single score, usually labeled y . There exists

some theoretical function, $F^*(\vec{x})$ that directly maps $\vec{x}$ onto y . This function cannot be known exactly for problems of moderate complexity, so an estimate of this function, $\hat{F}(\vec{x})$, is the goal of the gradient boosting process.

To quantify how closely the approximation $\hat{F}(\vec{x})$ follows the true $F^*(\vec{x})$, some loss function must be defined. There are many forms of loss function, as will be described in section 4.2. The loss function is termed $L(y, F(\vec{x}))$, meaning that it quantifies how closely the approximate points, $F(\vec{x})$, follow the true classification values of the events, y . Given these definitions, the goal of the BDT algorithm can be written as given by Friedman [6].

$$F^* = \min_F E_{y, \vec{x}}[L(y, F(\vec{x}))] \quad (4.1)$$

4.2 The Objective Function

4.3 XGBoost Software

Chapter 5

Development of an MVA-Based High Level Trigger

5.1 Limitations and Benefits of Current $H \rightarrow \gamma\gamma$ Trigger

As discussed in section 3.3, the current $H \rightarrow \gamma\gamma$ HLT uses hard cuts on kinematic, isolation, and showershape variables to select events. While cuts on some of these variables – such as R_9 and H/E – serve to remove low quality photon candidates, others reduce the phase space of the trigger while not necessarily increasing the quality of the analysis. Principally, the current HLT makes rather tight cuts on the p_T of the two photon candidates. These cuts significantly reduce the trigger rate, as many background processes produce showers with low p_T . Many signal events, however, have at least one candidate with low p_T . It will be shown in section 5.3 that these p_T cuts significantly decrease the acceptance of simulated Monte Carlo signal events. Since many future analyses will be searching for either very rare processes or physics beyond our current understanding of the standard model, constraining the phase space of available information may affect the collaboration’s ability to probe novel processes.

The relative simplicity of the standard HLT, however, provides multiple benefits. Firstly, the numerical cuts can be closely examined and understood very easily. Though the cuts have been relatively stable in time, they may adjusted and reexamined with relative ease. The effects of cuts on individual variables can be quantified, which allows maximum transparency into how a decision is made on an event-by-event basis. Secondly, a decision for each event can be made

very rapidly. As discussed in section ??, minimizing computation time for each event is vital given the huge number of triggers deployed at the trigger farm. Since all of the quantities used by the current HLT are already being saved for each event, there is very little additional computation needed to arrive at a triggering decision. Any proposed changes to the $H \rightarrow \gamma\gamma$ trigger must have minimal impact on decision speed given the stringent restrictions on computation time.

5.2 Development of a Boosted Decision Tree for Photon Candidate Classification

The HLT is a binary classification problem, where the two classes are signal events to keep and background events to discard. While the HLT is a per-event selection, $H \rightarrow \gamma\gamma$ events contain two photon candidates. As described in 3.3, the current HLT first makes requirements on individual photons, then applies a final filter that considers both photons together. Our MVA-HLT uses a similar philosophy; a binary classifier can be used to calculate an identification score for each photon, then combined cuts can be made on these scores for two photon candidates. The process of creating the MVA-HLT, then, is twofold. Firstly, the single photon binary classifier must be constructed such that signal-like photon candidates are given a high score and background-like photon candidates are given a low score. Secondly, cuts on both photon candidate scores must be selected such that the signal efficiency is maximized while the data rate is minimized.

5.2.1 XGBoost-Based BDT Software

XGBoost is a commonly-used software package throughout CMS, as it excels at binary classification problems. Specifically, Boosted Decision Trees (BDTs) are used commonly in CMS software (REF?). BDTs provide a good balance between model complexity and transparency. By tuning a relatively small number of model hyperparameters, as discussed in section 5.2.4, model complexity can be controlled and understood. For the development of this BDT, the python version of XGBoost was used, though the final implementation of the model must be done in the

C-based CMSSW (as discussed in section 5.4).

5.2.2 Signal and Background Samples for Classification

The first – and perhaps most important – step in creating a binary classifier is selecting samples on which to train the BDT. It is important that the signal sample chosen accurately represents the type of events that should be kept at the HLT, while the background sample must encompass all possible source of background events.

Signal Samples

As discussed in (REF?), Monte Carlo simulations exist for various physics processes that contribute to $H \rightarrow \gamma\gamma$. While there are several common production modes for the Higgs boson (as discussed in section (REF?)), gluon fusion Higgs production (ggH) has the largest production cross section. In addition to the large cross section, ggH production has fewer associated productions when compared to VBF or ttH Higgs production. For these reasons, ggH signal MC provides a clean signal that maximizes the photon discrimination ability of a BDT.

In order to validate the performance of the model, the acceptance of these signal MC events must be quantified. As a result, half of the sample is used for training, while the other half is reserved for later validation. Events are split by event number into two separate categories: even event numbers are used for the training sample, and odd event numbers will be used for the validation of the model performance.

Background Samples

At the trigger level, background simulation poses a major challenge. Background events for $H \rightarrow \gamma\gamma$ fall into three major categories: events containing two photons diphoton, events containing a photon and a jet, and events containing two jets. While it is possible to simulate these background sources, the simulation – especially of the jet-jet events – is challenging and therefore may not perfectly align with the real data. The most obvious resolution to this issue is to simply treat the data sample as background. While, of course, the goal of the trigger is to

Table 5.1. Table of input variable names, and a short description of each.

Variable Name	Description
Raw Energy	The total energy deposited in the 5x5 supercluster
R_9	The proportion of the energy contained in the central 3x3 crystals relative to the raw energy
$\sigma_{i\eta i\eta}$	The covariance in terms of crystal spacing (i_η)
η_{SC}	The eta coordinate of the center of the supercluster
η Width	The supercluster width in η
ϕ Width	The supercluster width in ϕ
H/E	The ratio of the total supercluster energy measured in the HCal to that measured in the ECal
ECalPFIso	The isolation energy of the particle deposited in the ECal

preserve signal signatures from the data, it has been shown in chapter ?? that the overall expected count of signal events in this data sample is relatively low. As long as a clean signal sample is provided to the BDT, the incredibly small percentage of possible $H \rightarrow \gamma\gamma$ events in the data should not appreciably mar the model’s ability to differentiate signal and background photon candidates. As with the signal MC, even event numbers are used for the training, while odd event numbers are reserved for later validation of model performance.

5.2.3 Determination of Input Variables

After signal and background samples are selected, input variables for the BDT must be chosen such that signal and background events can be separated. The most obvious starting place for this is by analogy to the current HLT; since cuts on the kinematic, isolation, and showershape variables have been successfully used in the preexisting trigger, a BDT should be able to use these variables to even greater effect. Below, in table 5.1, the chosen input variables are listed along with their definitions.

The chosen variables are largely showershape and isolation variables, as these can be used to differentiate a legitimate photon signature from a jet shower. A major goal of the BDT is to increase kinematic coverage; relying primarily on showershape variables ensures that the selection is not biased towards a specific kinematic region. Below are plots of the above input

variables, where the signal MC is shown in black and the data (background) is shown in red.

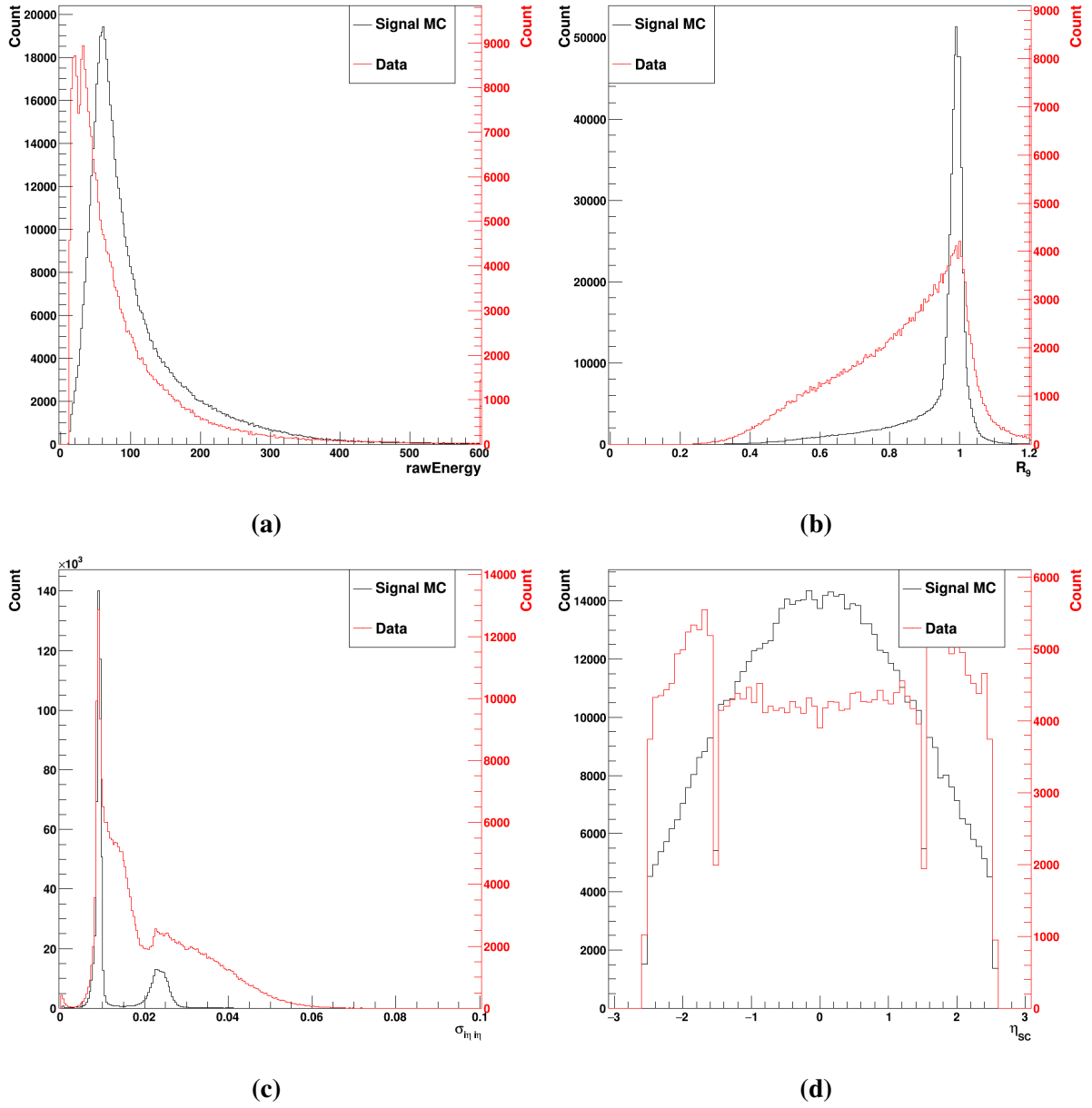


Figure 5.1. Plots of the input variables to the BDT, where the signal MC is shown in black and the data is shown in red.

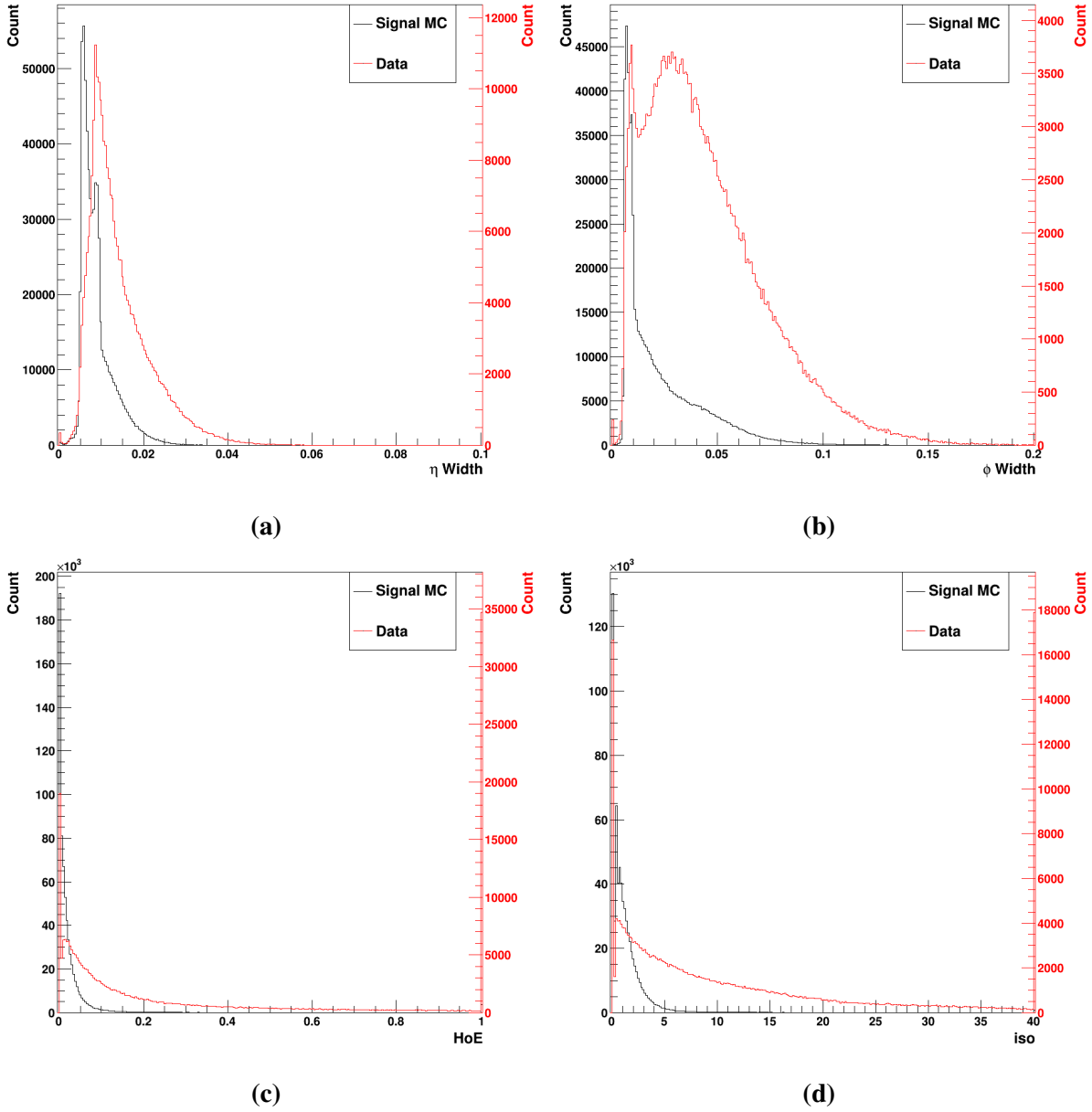


Figure 5.2. Plots of the input variables to the BDT, where the signal MC is shown in black and the data is shown in red.

5.2.4 Hyperparameter Tuning

Once the input samples and variables are determined, the technical details of the model must be optimized. The structure of the BDTs is determined by the hyperparameters of the model as discussed in chapter 4. The most fundamental hyperparameters are the maximum depth, the learning rate, and the total number of trees in the model. The max depth and number of trees

directly determine the complexity of the model. A model with more trees will, based on the boosted gradient descent method discussed in chapter 4, more precisely estimate the function that maps the chosen input variables from the training sample onto the correct labels of signal versus background. To mitigate the risks of overtraining the model, XGBoost early stopping is used to determine the number of trees. As the software adds additional trees to the ensemble, it measures the chosen objective function at each step on a testing sample not used in the training. Early stopping ends the training of the model when the chosen metric hasn't improved in a chosen number of rounds. For this training, 50 early stopping rounds were used and the area under the curve (AUC) was used as the target metric. So, then, this hyperparameter is essentially fixed while the others must be optimized.

The maximum depth controls the complexity of each individual tree in the ensemble. As discussed in chapter 4, the depth determines the number of layers each tree is able to employ. Deeper trees generally allow more precise mapping from input values to output scores at the cost of computation time; each layer of depth constitutes another set of weights that must be computed for each tree. Since it is vital that the computation time is minimized at the trigger level, this is a major consideration. Deeper trees also tend to overfit the training data more, as there are essentially more degrees of freedom with which to fit the mapping function. So, the max depth must be kept as low as possible to avoid both overfitting and unreasonable computation time.

The learning rate controls the size of the individual feature weights from each new tree added to the ensemble. A higher learning rate allows each tree to have higher weights, which will therefore affect the model performance more drastically. Lowering the learning rate makes the model more conservative, as it reduces the overall change to the model predictions provided by each additional tree. Having a higher learning rate generally reduces the number of trees necessary to converge, but decreases the accuracy since statistical fluctuations within each tree can be assigned higher weights.

It was found that the max depth and learning rate are best optimized in tandem; as the

trees become more complex (higher max depth), it was found that making the individual steps more conservative (lower learning rate) led to less overfitting and higher performance. As a result, a 2D grid search was used to optimize these two hyperparameters. A range of both max depth and learning rate was defined, then a model was trained for each combination. The resulting models were appraised in terms of not only their objective scores (AUC and logistic loss in this case, both described in chapter 4), but also in terms of the number of trees and the overall model size in megabytes. Since the model must be loaded into RAM at the trigger farm and then the weights applied to each photon in each incoming data event, a model taking up more space will increase both the computation time and memory load at the trigger level.

After many trials, the best balance of model performance, minimization of overfitting, and model size was found to be a max depth of 7 and a learning rate of 0.25. Experimentally, it was determined that many of the other hyperparameters had minimal impact on the model performance and size, so they were therefore left at their default values.

5.3 Determination of BDT Cuts

After the tuning and training of the model was optimized, a score is generated for each photon in each event. Each event may contain several photon candidates, but the job of the HLT is to determine if any pair of candidates exists that plausibly originates from $H \rightarrow \gamma\gamma$. Given that the BDT was trained such that photons originating from $H \rightarrow \gamma\gamma$ MC are assigned a high score and background processes are given a low score, the photon candidates are first sorted by the output BDT score. Then, the highest two scores can be selected and the values analyzed for both signal MC and data events.

Since the HLT path simply renders a decision about whether an event contains a $H \rightarrow \gamma\gamma$ process, cuts on the pairs of output scores must be chosen such that the number of signal events kept is maximized (high signal efficiency) while the total number of data events kept is minimized (low trigger rate).

Below, in figure 5.3, the high and low scores for signal MC are shown at left. Since these are purely signal events, it is expected that the scores to be clustered near 1. As can be seen, the higher of the two scores (along the x axis) tends to be above 0.9 for a large percentage of events. Meanwhile, while the lower of the two scores does cluster above 0.8, there is still a large band of score values extending down to 0. This means that there are many signal events for which one photon has a very high score, but the second photon has a lower score.

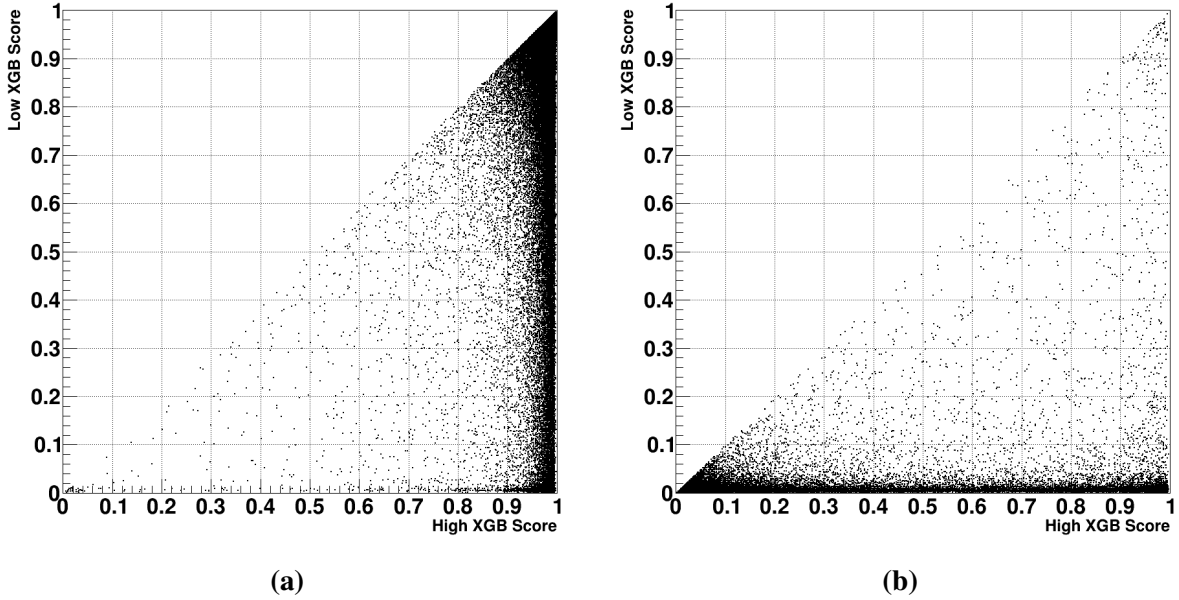


Figure 5.3. XGBoost output scores for two selected photon candidates in the Winter 2024 ggH signal MC. The higher of the two scores is plotted on the X-axis, while the lower of the two is plotted on the Y-axis.

Next, consider the scores of the data sample shown at right in in 5.3. Given the rarity of $H \rightarrow \gamma\gamma$ events at the trigger level, it is expected that essentially all of these data events correspond to background processes. As a result, the vast majority of these data photon candidates should be assigned a BDT score close to zero. As can be see, the lower of the two photon scores very rarely extends above 0.1. Though the higher of the two scores is often close to 0, there are still many events where the value extends towards 1. Since this BDT is designed to indentify individual photon candidates, it can be inferred that many of these data events have one real photon (which is a major source of background as discussed in section 5.2.2).

As discussed in section 5.2.1, the single photon BDTs are trained separately for photon candidates in the barrel and those in the endcap. Due to the separate trainings – as well as the differences in kinematics and detector geometry – the score distributions for barrel and endcap photon candidates are not perfectly equivalent. As a result, the chosen cuts must be tuned separately for the barrel and endcap photons. This presents an issue, as diphoton events may contain two barrel photon candidates (BB), two endcap photon candidates (EE), or one photon candidate in each region of the detector (BE). The simplest way to deal with this is to choose the single photon cuts by reference to the BB and EE distributions, then apply the chosen barrel and endcap cuts to all photons in either region.

Given the shapes of the output distributions in figure 5.3, it is clear that a single cut value for both photons will not achieve maximum signal efficiency. While many of the high score photons are clustered above 0.9, there are still many events with a second photon candidate that has a lower score. In order to preserve these signal events while still removing much of the background, cuts can be selected such that the cut value on the lower score depends on the higher score. For first photons with a very high score, a very loose cut can be applied to the second photon. As the score of the first photon decreases, the cuts on the second photon can be tightened. In the case where the first photon has a relatively low score, a tighter cut is applied on the second photon to help reduce the number of events with two bad photon candidates in the data while still preserving much of the signal. Since the data at the trigger level is strongly dominated by di-jet or single photon events, these cuts can drastically reduce the data rate while still preserving much of the signal. These cuts were experimentally determined, and are tabulated in table 5.2.

Table 5.2. Chosen score cuts for BB and EE events

(a) Score cuts for BB events		(b) Score cuts for EE events	
High Score	Minimum Low Score	High Score	Minimum Low Score
0.92	0.02	0.95	0.04
0.85	0.04	0.85	0.08
0.30	0.14	0.5	0.20

Events that satisfy these requirements can be plotted on top of figures 5.3 . In figure 5.4 and 5.5 below, the selected events for BB and EE can be seen, respectively, in red. Yellow lines are added to the plots to demarcate the cuts. These plots contain both the signal MC (left) and the Data (right).

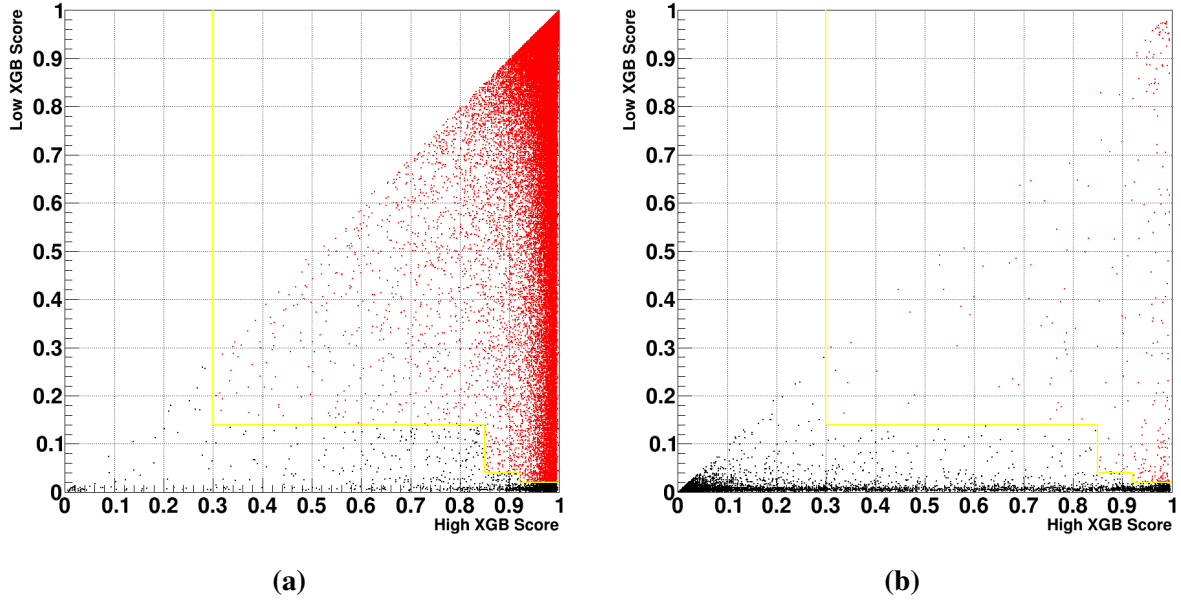


Figure 5.4. XGBoost output scores for two selected photon candidates in the Winter 2024 ggH signal MC (a), and 2023 Data (b). The higher of the two scores is plotted on the X-axis, while the lower of the two is plotted on the Y-axis.

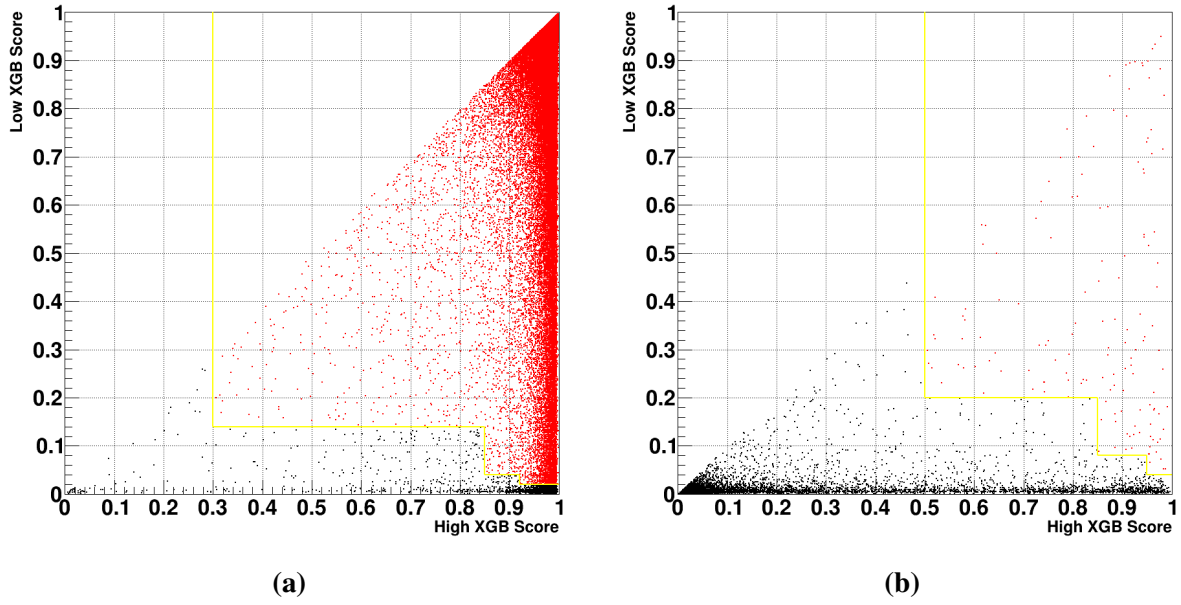
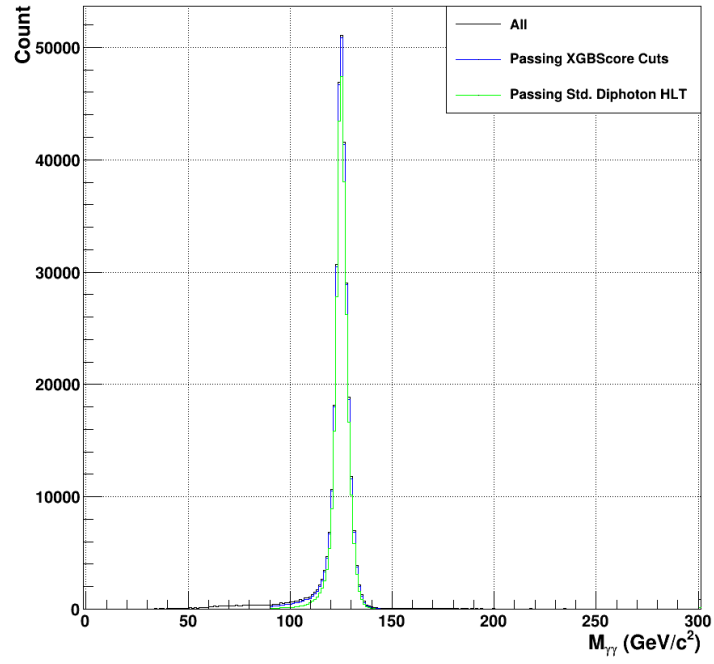


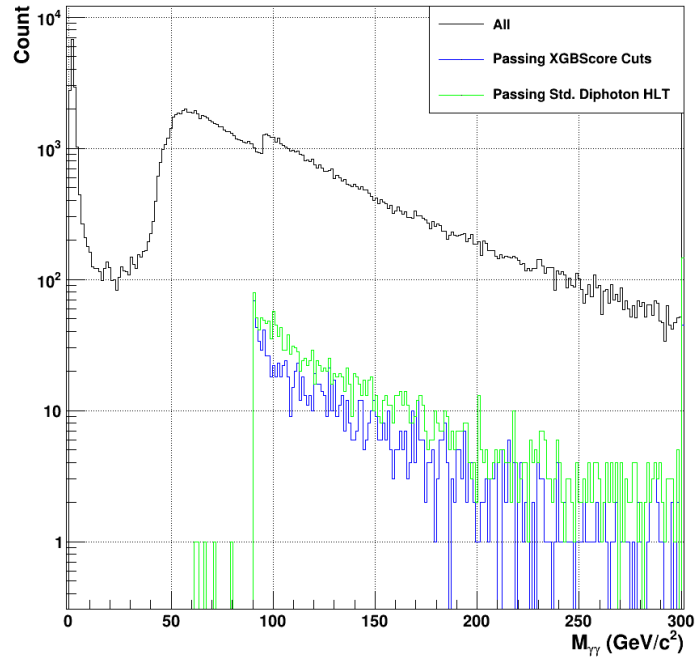
Figure 5.5. XGBoost output scores for two selected photon candidates in the Winter 2024 ggH signal MC (a), and 2023 Data (b). The higher of the two scores is plotted on the X-axis, while the lower of the two is plotted on the Y-axis.

This choice of cuts, clearly, is able to preserve much of the signal MC while accepting very few of the data events examined. While the overall signal efficiency and data rate are key metrics to appraise our MVA-HLT, it is also important to examine where in the kinematic phase space events are being gained relative to the standard diphoton HLT. While a more complete measurement of this will be shown in chapter 7, a preliminary analysis can be made with the cuts applied in this initial development framework.

Below, plots of selected kinematic variables are shown. The events accepted by the current diphoton HLT are shown in green, while the events accepted by the new MVA-HLT are shown in blue. As with the previous score plots, the Winter24 GGH signal MC will be shown at left, whereas the 2023 Data will be shown at right.



(a)



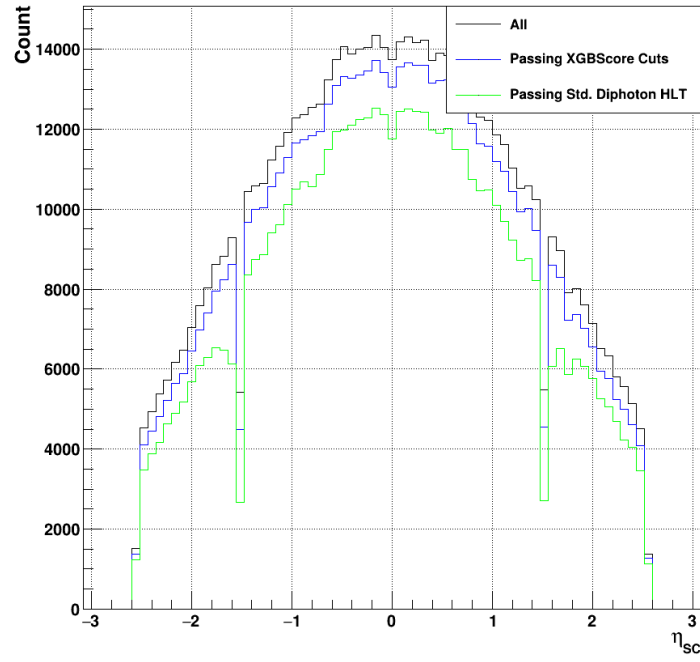
(b)

Figure 5.6. Diphoton mass for the signal MC (a) and data (b). Events accepted by the MVA-HLT are shown in blue, while events accepted by the standard HLT are shown in green.

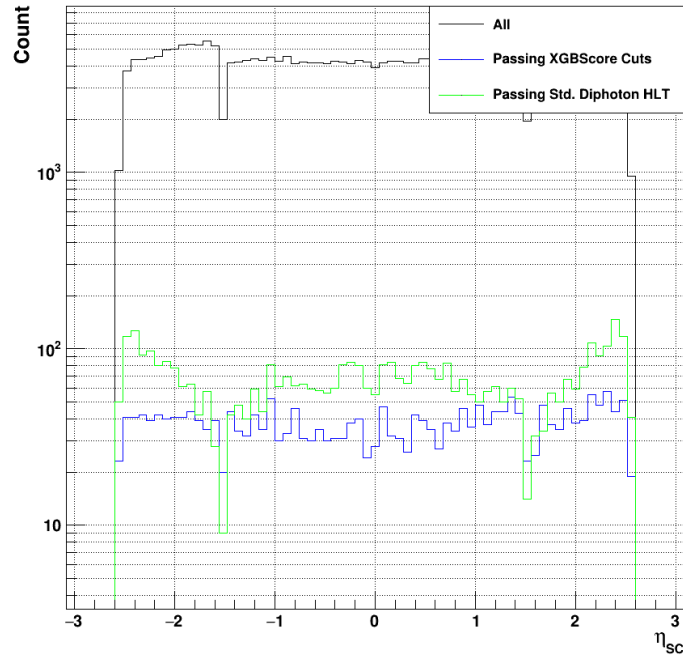
Beginning with the mass, it can immediately be seen that MVA-HLT accepts 94.2% of signal MC events, which is an approximately 13.6% relative gain in signal efficiency compared to the 82.9% acceptance of the standard diphoton HLT. Additionally, the data rate appears considerably lower; while the standard HLT accepts 1.44% of the analyzed events, the MVA-HLT only accepts 0.86%, which corresponds to a 40.3% reduction in data rate.

One major consideration when examining the mass is the measured mass width of the accepted signal events. Generally, a more narrow peak is considered a cleaner signal. By treating the mass distribution as approximately gaussian, the width can be easily understood by measuring the standard deviation. Since the distribution is only approximately gaussian, it is safer to instead construct the quantity $2\sigma_{effective}$, which is simply the width (in GeV/c^2) that contains 95% of the peak. Although this is not a statistically rigorous construct, it can be used to compare the approximate shapes of similar distributions in an internally consistent way. It can be seen that the $2\sigma_{effective}$ is marginally increased for the MVA-HLT, reaching 6.58 compared to the 6.07 of the events accepted by the standard trigger. Some increase in width is to be expected, as accepting more events will necessarily bring the distribution closer to the "all" curve that has a wider width than either selection.

(Statments about the background mass distribution?)



(a)

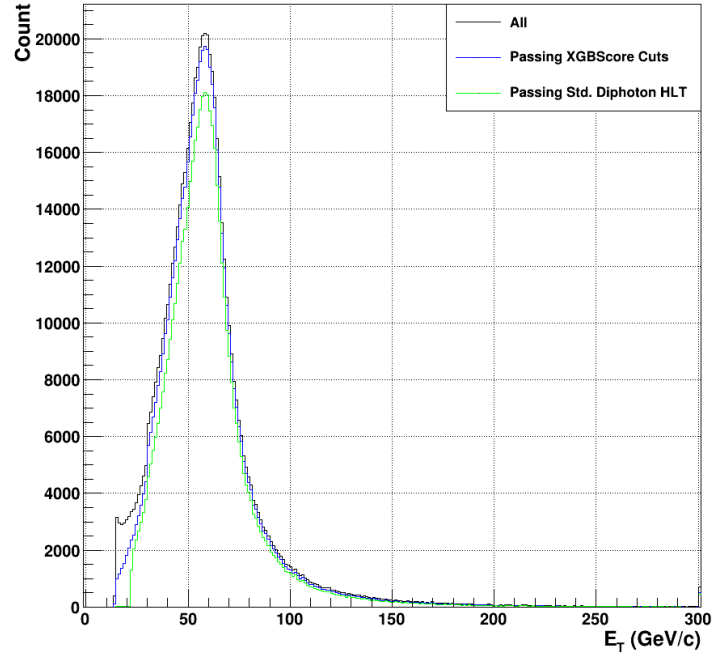


(b)

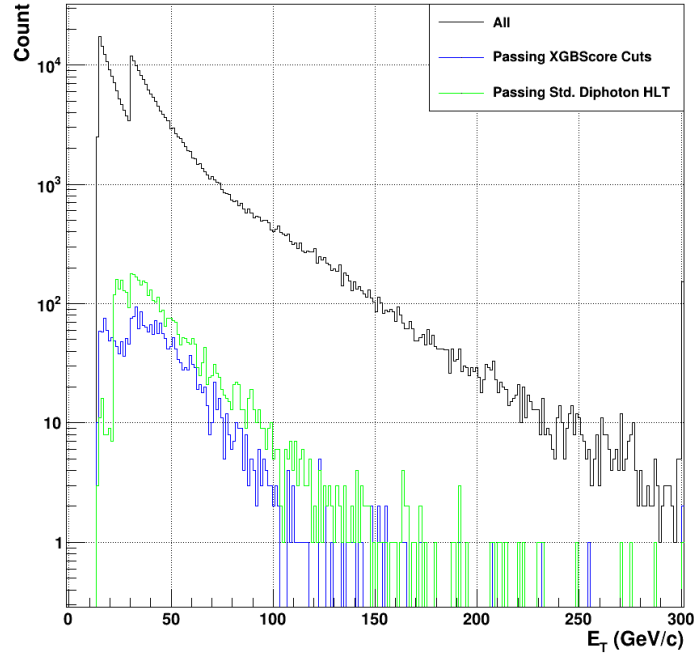
Figure 5.7. Photon η for the signal MC (left) and data (right). Events accepted by the MVA-HLT are shown in blue, while events accepted by the standard HLT are shown in green.

Above, the next kinematic variable shown is the η_{sc} of both photons plotted together. As seen in both signal and background, the separation between the ECal Barrel (EB) and ECal Endcap (EE) can be seen at $\eta_{sc} \approx \pm 1.5$. For the signal, throughout the barrel, both the MVA-HLT and standard HLT accept photon candidates in relatively constant proportion to the total distribution, though the MVA-HLT has overall improved acceptance. While the MVA-HLT still has relatively constant acceptance in the EE, it can immediately be seen that the standard HLT is excluding many events in the lowest- η_{sc} region of the endcap. Investigations have shown that this is due to the R_9 cuts made within the trigger. This represents a reduction in kinematic coverage for the standard HLT, whereas the MVA-HLT is able to resolve this issue and maintain maximum kinematic coverage.

Considering the background, it can immediately be seen that the MVA-HLT has a relatively flat acceptance in both the EB and EE, while the standard HLT appears to accept more data at the extreme- η_{sc} regions of the endcap. As can be seen from the MC, this region tends to contain less of the $H \rightarrow \gamma\gamma$ signal events and therefore represents a lower signal to background ratio.



(a)



(b)

Figure 5.8. Photon E_T for the signal MC (left) and data (right). Events accepted by the MVA-HLT are shown in blue, while events accepted by the standard HLT are shown in green.

Finally, consider the transverse energy of the photon candidates, E_T . Here, both candidates are again shown on the same plot, which explains the double-peaking behavior in the background distribution. The standard diphoton HLT contains cuts at 30 and 22 GeV/c^2 for the lead and sublead photon, respectively. The sublead cut can clearly be seen in both the signal, at left, and the background at right. While the new MVA-HLT selection accepts more of these low- E_T photons, it is clear that little of the gain comes from this kinematic region. Below about 30 GeV/c^2 , both selections tend to cut out many photons from the total distribution. In the case of the MVA-HLT, this is likely due to those photon candidates being of lower quality and therefore being assigned a lower BDT score.

Below, in table 5.3, the acceptance percentages are given for the 3 different diphoton η_{sc} configurations, as well as the total acceptance for both signal and background, compared to the standard HLT cuts.

Table 5.3. Signal efficiencies and background acceptances for both triggers, broken down by diphoton η_{sc}

(a) Signal Efficiencies			(b) Data Acceptances		
Region	MVA-HLT(%)	Std. HLT(%)	Region	MVA-HLT(%)	Std. HLT(%)
B-B	95.93	89.32	B-B	0.65	1.06
B-E	91.91	74.10	B-E	1.20	2.17
E-E	89.46	76.85	E-E	0.58	1.07
All	93.86	82.64	All	0.88	1.56

5.4 Final Implementation in HLT Menu

As discussed in section 3.3.3, triggering and event reconstruction is handled by the distributed CMSSW software framework. As a result, any proposed trigger developments must be deployed in a central CMSSW release. While the software strategy is the same as the original python implementation, the code must of course be rewritten in C and adapted to work within the trigger.

As described in section 3.3, the HLT operates on all photon candidates in an event, then

simply renders an event-by-event decision if any pair of photons satisfy the chosen conditions. As a result, the first step of the implementation is to create a module that calculates the BDT score for each photon candidate in every event. Once these scores are stored within the trigger path, another module must be constructed to check if any possible pair of photon candidates passes the two-fold cuts described above. Importantly, due to the constraints on storage space, the scores of the individual photon candidates cannot be saved to the event. As a result, the only output from the MVA-HLT is a decision on whether the event should be accepted or rejected.

Luckily, these two modules are the only software additions required to add the BDT to the trigger menu. Beyond that, standard modules can be used (with some modifications to the cuts made) within ConfDB to construct the entire trigger path as discussed in section ?? . The first step of the HLT path is to require the necessary L1 triggers. The standard L1 suite required by the existing HLT is used here, which consists of a combination of single and double EGamma triggers as described in section 3.1.

In order to reduce the amount of MVA scores that must be calculated, the very loose kinematic cuts are applied to all photon candidates in the event as the first step. While these cuts are quite loose, they still serve to reduce the number of candidates considered. Since many data events may have many photon candidates, this will help reduce the time taken to calculate the BDT scores. (Is this true? Do individual photon candidates get removed at the HLT level, or just events?)

With the preliminary cuts in place, the module that calculates the scores is placed in the HLT path, followed by the module that makes the cuts as described in section . Finally, a combined module is created that combines these BDT cuts on both photons with a mass cut of $M_{\gamma\gamma} > 90 \text{ GeV}/c^2$. So, then, the whole HLT path serves to check if there is any possible pair of photon candidates that passes the loose kinematic cuts, the pairwise BDT cuts, and the $M_{\gamma\gamma}$ cut. If any pair satisfies these conditions, a flag is added to the event and all photon candidates are kept at the RAW level.

Chapter 6

MVA-Based Offline Preselection

As discussed in chapter 3, the HLT is one of the first steps in selecting data and MC events for analysis. Once a decision has been rendered by the HLT, the event is read out and saved in either the appropriate data stream or in the corresponding MC sample. Once these files are read in by the analysis software – in this case HiggsDNA as described in section ?? – further selection cuts are made to create a final analysis sample on which measurements can be made.

6.1 Current Diphoton Preselection

The current diphoton selection is created by direct analogy to the standard diphoton HLT; firstly, p_T , showershape, and isolation cuts are made on individual photons in an event, then the single photons are paired together and cuts are made on the resulting diphoton. In the case of the HLT, a simple diphoton mass cut is made. At the offline analysis level, however, the diphoton cuts are made on $p_T/M_{\gamma\gamma}$.

The $H \rightarrow \gamma\gamma$ preselection is described in detail in [12]. First, loose cuts of $p_T > 20\text{GeV}/c$, $R_9 > 0.5$, $H/E > 0.08$, and $idMVA > 0.9$ are applied to each photon candidate. Additionally, the $|\eta_{sc}|$ is constrained to be below 2.5, and the gap region between the EB and EE is excluded ($1.444 < |\eta_{sc}| < 1.556$). Additionally, the electron veto is required to exclude photons originating from electron tracks. Beyond that, the remaining isolation and showershape cuts are only applied on low- η photons. Above $R_9 = ??$ in the barrel and $R_9 = ??$ in the endcap, no further preselection

cuts are applied. For photons below this R_9 , cuts are applied on the $\sigma_{i\eta i\eta}$, –

Once these single photon preselection cuts are applied, all remaining photon candidates are paired together into a collection of diphotons. For each pair, the higher of the two photons must satisfy $p_T > 30\text{GeV}/c$. Paired with the baseline requirement of $p_T > 20\text{GeV}/c$, these cuts closely mimic the 30 and 22 GeV/c cuts applied at the trigger level. Then, the $p_T/M_{\gamma\gamma}$ cuts are applied to the diphoton pairs.

At this point, many of the diphoton pairs have been removed. A final diphoton combination must be selected, however, as HiggsDNA is designed to output a single photon pair of photons. The remaining pairs in the event are sorted by their diphoton p_T , and the pair with the highest value is written out into the event.

There is no explicit $M_{\gamma\gamma}$ cut applied within the preselection, as the possibility for high- and low-mass Higgs analysis is left open. For the purposes of comparing this preselection to the new MVA-based selection, only the standard mass Higgs is considered. So, a loose cut of $90 < M_{\gamma\gamma} < 180\text{GeV}/c^2$ is applied to all resulting diphotons.

6.2 Development of MVA-Based Offline Preselection

Since the offline selection is designed to cover the HLT used, it is clear that applying the preexisting preselection described above on top of the MVA-HLT is a suboptimal strategy. In fact, since these cuts mirror the standard HLT quite closely, it is expected that applying the standard preselection on top of the MVA-HLT would likely reduce or even entirely remove the signal efficiency gains obtained by the new trigger. As a result, an alternative set of preselection cuts must be proposed that allows the trigger-level gains to be realized at the offline analysis level.

Just as the standard preselection cuts are designed to not only cover the standard HLT but work in much the same way, a new preselection can be created with maximum similarity to the MVA-HLT. One major complication of this is that the same BDTs cannot be used for both the

trigger and offline selection. Since the reconstruction differs for the NanoAOD vs the RAW, the input variable distributions will be different enough that the models are not directly comparable. Instead, a new BDT-based preselection framework must be developed and its coverage of the MVA-HLT must be analyzed.

6.2.1 Creation of Single Photon BDTs

A $\gamma + Jet$ BDT

As discussed in section 6.1, there already exists an MVA within the standard preselection. While this idMVA is now a centralized part of the NanoAOD format, historical preselections used an MVA trained on $\gamma + Jet$ MC (REF?). Events in $\gamma + Jet$ MCs will contain one real (prompt) photon and one jet that looks like a photon, termed a "fake". This MC provides a very easy-to-use training sample, as one can simply use the prompt γ as the signal photon and the fake as the background photon. This standard $\gamma + Jet$ idMVA is usually only used to make relatively loose cuts as part of the much broader preselection cuts as described above.

The original idMVA was trained using many of the variables from the preselection as input variables. These same input variables were used as a starting point for a new $\gamma + Jet$ MVA, with the addition of H/E. By adding this variable to the inputs, the new $\gamma + Jet$ MVA contains the same information available to the standard preselection described above. Additionally, this allows the offline selection model to more closely mirror the MVA-HLT. The input variables used are shown below in table 6.1. As with the MVA-HLT, the barrel and endcap models are trained separately, with individual photons selected based on their location in the detector.

As with the trigger level BDT, the separation between the signal and background in the various input variables can be examined. For the subsequent plots (6.1-6.7), the prompt and fake will be shown for both the barrel and endcap, as the trainings are separated. The prompt photons will be shown in dark and light blue for the barrel and endcap, respectively. The fake distributions will be shown in red and pink for the barrel and endcap, respectively. All the input variable distributions have been normalized to 1, so that their shapes can be directly compared.

Table 6.1. Table of input variable names for the offline $\gamma + Jet$ MVA, and a short description of each. The final two variables are only used for training the endcap.

Variable Name	Description
Raw Energy	The total energy deposited in the 5x5 supercluster
R_9	The proportion of the energy contained in the central 3x3 crystals relative to the raw energy
S_4	The proportion of the energy contained in the central 2x2 crystals relative to the raw energy
$\sigma_{i\eta i\eta}$	The covariance in terms of crystal spacing (i_η)
$cov(i\eta i\phi)$	The covariance between the η and ϕ coordinates
η_{SC}	The eta coordinate of the center of the supercluster
η Width	The supercluster width in η
ϕ Width	The supercluster width in ϕ
H/E	The ratio of the total supercluster energy measured in the HCal to that measured in the ECal
PhoIso03	The photon isolation
Charge Iso w.r.t. Worst Vertex	The charged isolation for the worst vertex determined by the vertexing algorithm
Charge Iso w.r.t. Chosen Vertex	The charged isolation for the chosen vertex determined by the vertexing algorithm
ρ	The event energy density
esEffSigmaRR	temp
esEnergyOverRawE	temp

The SC raw energy, in 6.1a, shows rather clear separation for both the barrel and endcap. It can be seen that the prompt photons tend, on average, to have higher total supercluster energy than the fake photons. The separation in η_{SC} , however, is primarily seen in the endcap, where the fake photons tend towards the higher values. Since the BDT works by creating correlations between variables, and the η_{SC} is a strongly kinematic-dependent variable, it will still be shown to be a relatively powerful discrimination variable.

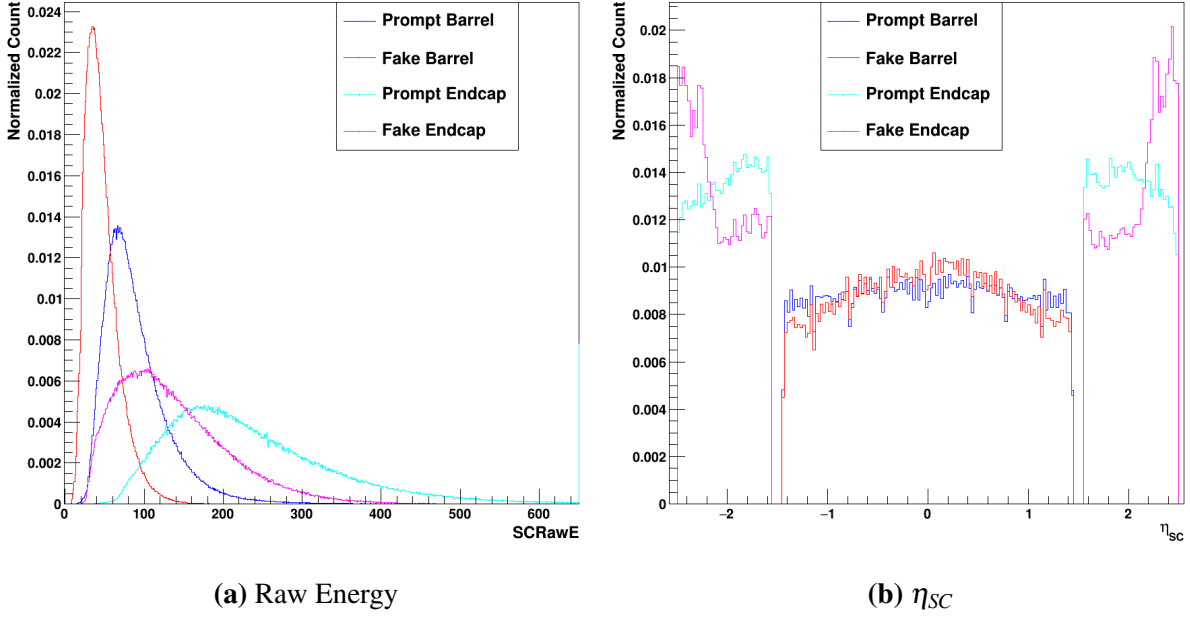


Figure 6.1. Plots of the Supercluster raw energy and η_{SC} . The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

Next, the energy ratios, R_9 and S_4 can be examined. It is expected that a real (prompt) photon will have relatively well-localized energy; since the photon is a relatively clean signal in the detector, it is expected that most of the energy will be deposited in the center of the supercluster. It is clear that this is the case for both the barrel and endcap, as seen in figure 6.2. Since the fake photon corresponds to a jet, it is expected that the energy deposition will be more spread in the supercluster, which can also be seen in the plots. As a result, R_9 and S_4 can be relatively powerful discrimination variables.

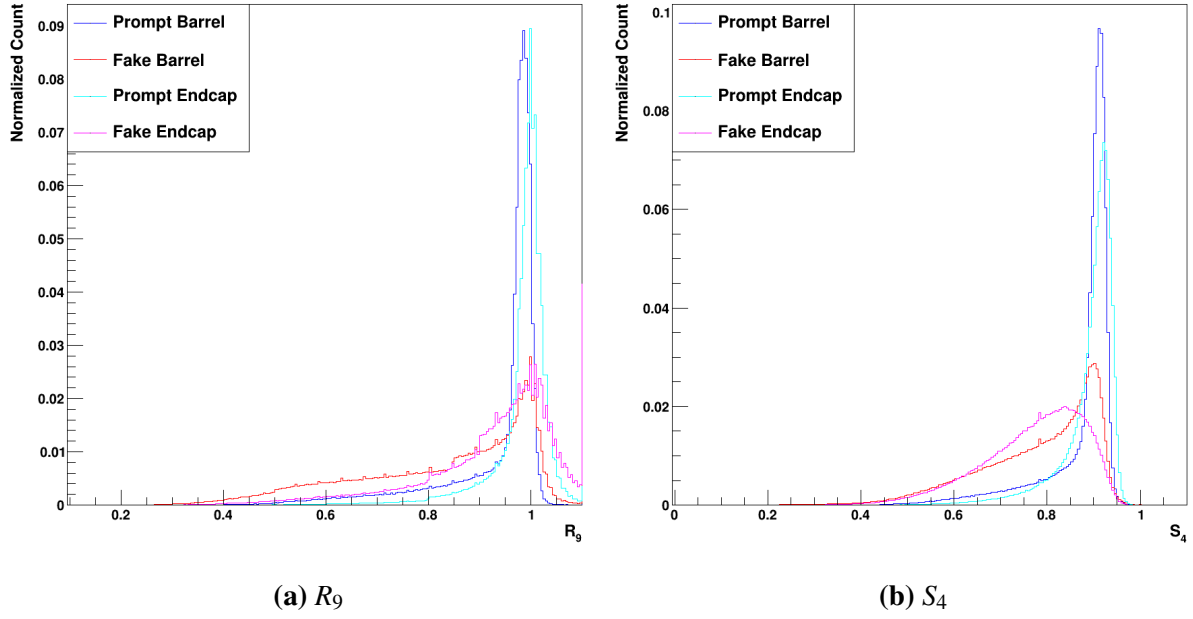


Figure 6.2. Plots of the R_9 and S_4 . The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

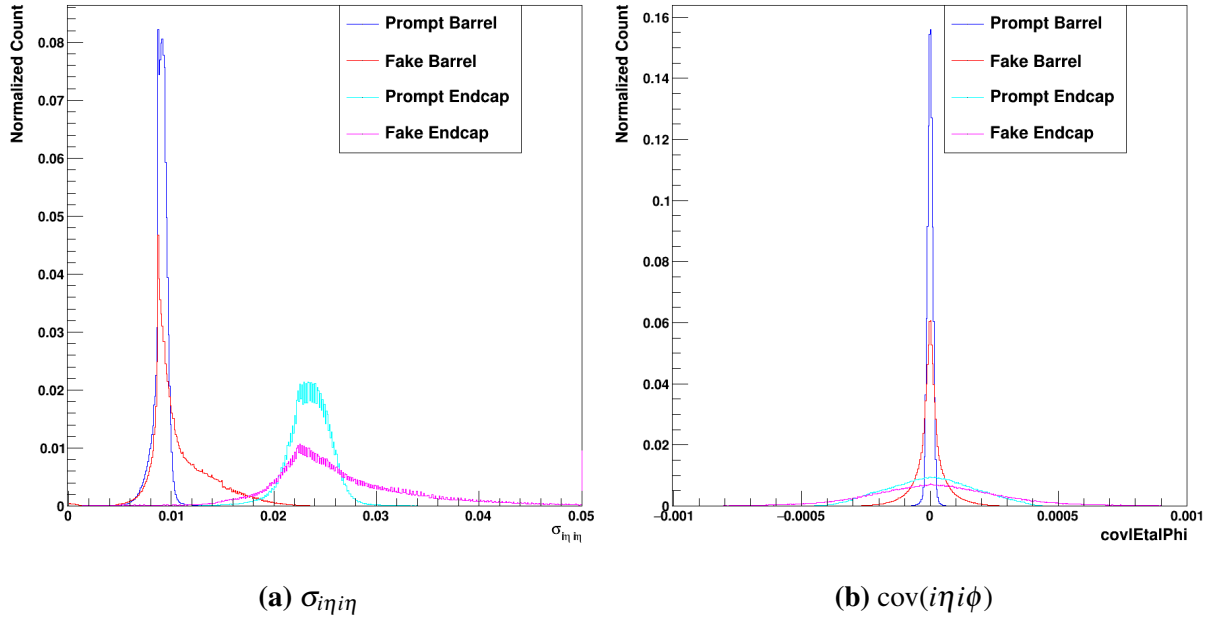


Figure 6.3. Plots of the $\sigma_{i\eta i\eta}$ and $\text{cov}(i\eta i\phi)$. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

Similarly to the R_9 and S_4 , the η and ϕ widths are powerful showershape variables that provide some measure of how well-localized the energy deposition is. It is expected that these values are different for the barrel and endcap, as the crystal size and spacing are different in the two different regions of the detector. This can be seen in both plots in figure 6.4. Additionally, the separation between prompt and fake photons can be clearly seen, especially in η width.

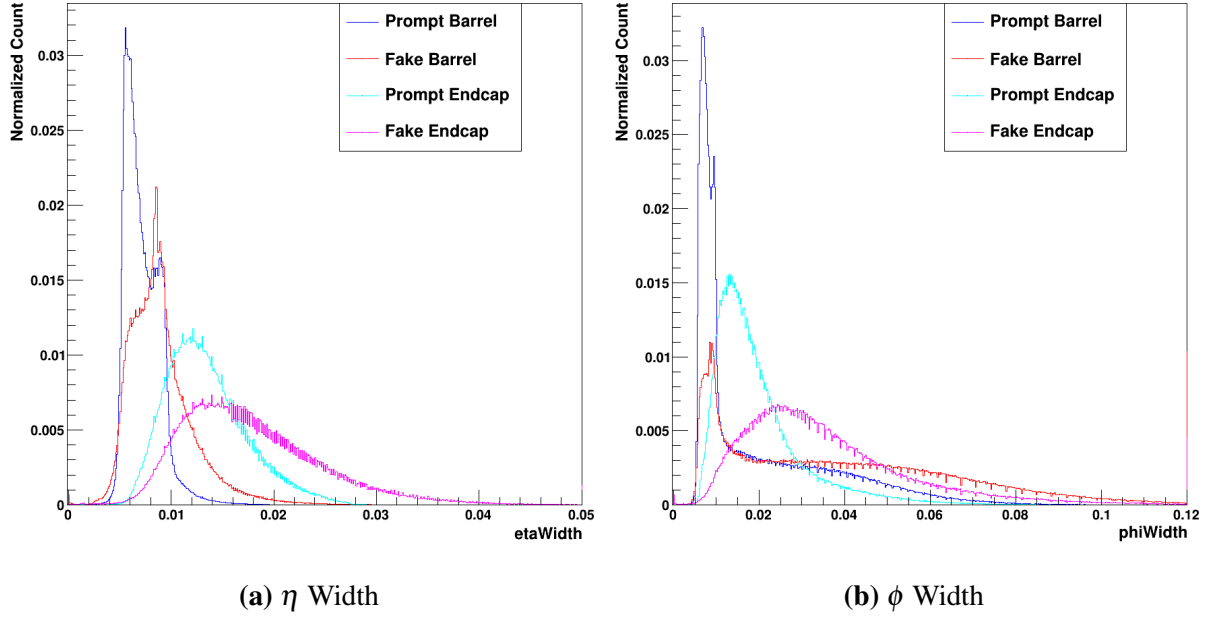


Figure 6.4. Plots of the η and ϕ widths. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

The measure of the HCal energy vs the ECal energy, H/E , provides a powerful measure of whether the energy deposition more closely matches an electromagnetic process – such as a photon or electron – or a hadronic process such as a jet. It is expected that most photons will have a value consistent with 0, as they should deposit all of their energy into the ECal. There is, of course, some level of energy leakage that ensures that even photons often do not have an H/E of identically 0. Clearly, however, the prompt photons in both the barrel and endcap strongly peak near 0 and fall off rather quickly below this. Note that figure a is a log plot, so that the secondary peak below 0 is down by more than a factor of 10 relative to the primary peak at 0.

Since the fake photons are jet signatures designed to mimic a prompt signature, they still exhibit a strong peak near 0. It can be seen, however, that the distribution extends to much higher values of H/E overall.

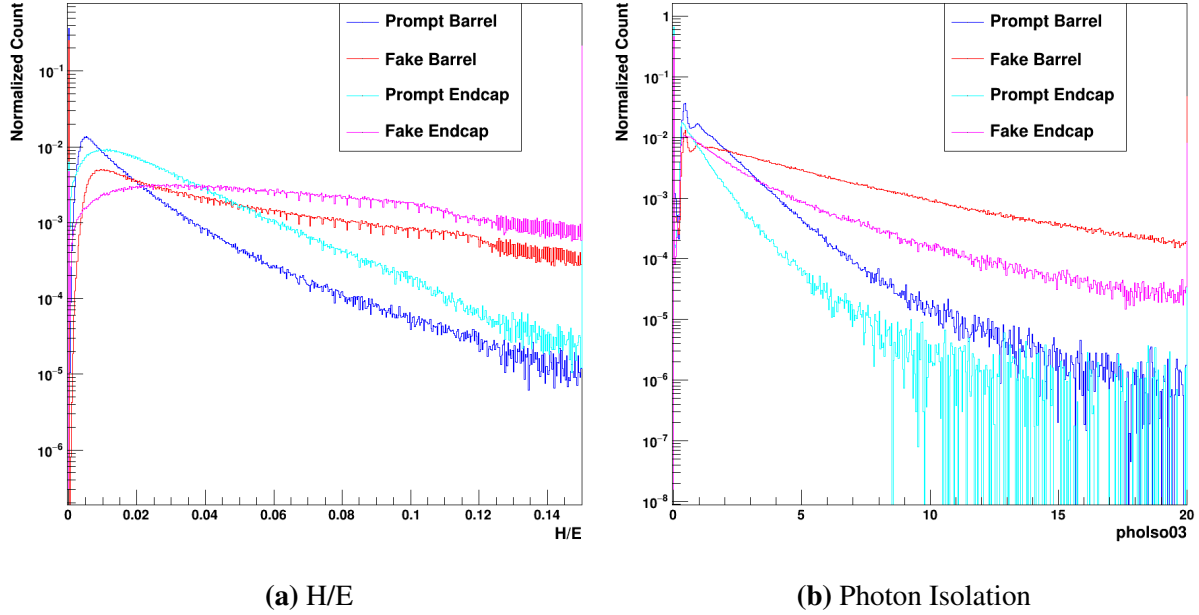


Figure 6.5. Plots of the H/E and photon isolation. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

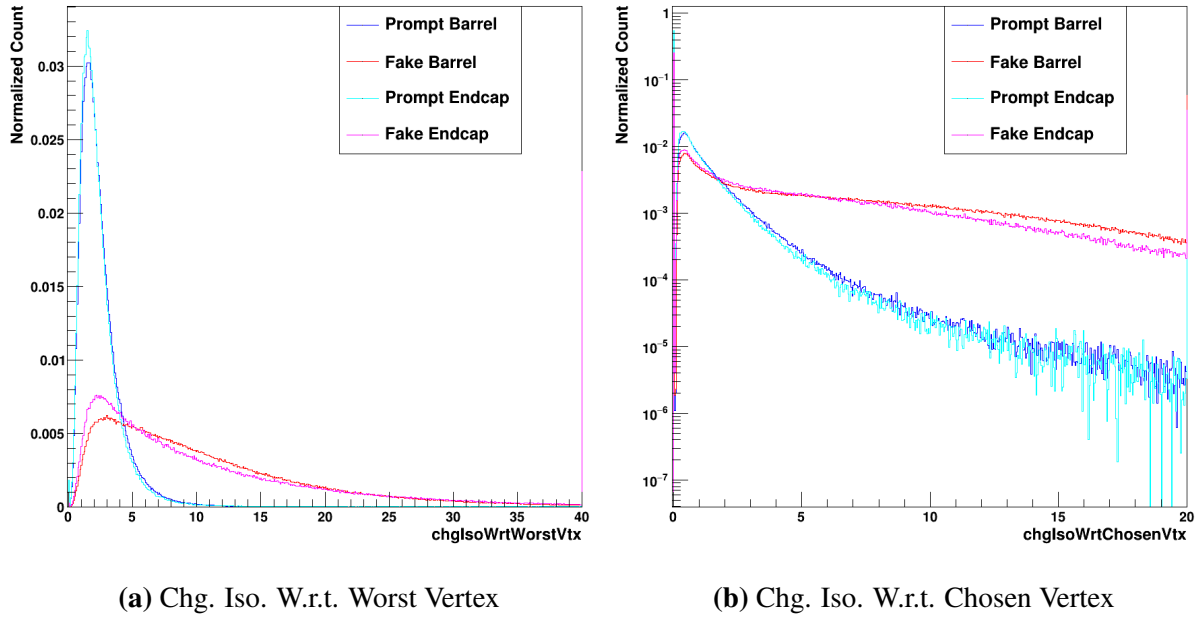


Figure 6.6. Plots of the charged isolations with respect to the chosen and worst vertices. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

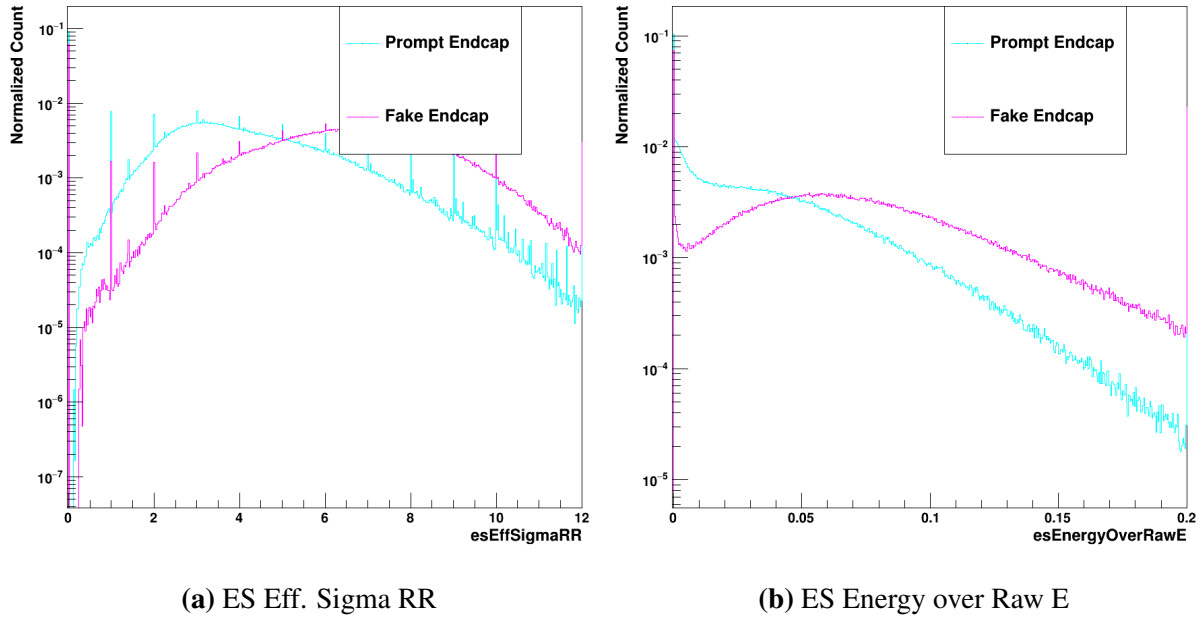


Figure 6.7. Plots of two endcap-specific variables: esEffSigmaRR and esEnergyOverRawE. The prompt signal is shown in light blue, whereas the fake background is shown in pink.

Comparing to the MVA-HLT variables shown in 5.1, it can be seen that a few variables

have been added at the offline selection level. The photon isolation is analogous to the ECalIso used at the trigger level, but is calculated differently (how?). The two charged isolations used are entirely new additions, as the MVA-HLT is calculated before any complex vertexing is implemented. (Why are both used?). (Statements on rho, and the endcap variables)

The hyperparameter optimization at the offline level was performed very similarly to the MVA-HLT; the max depth and learning rate were varied in tandem until an optimal point was found. The same process of early stopping within XGBoost was used, as well. The hyperparameters were, however, found to be quite different from the trigger-level optimization. Since there are considerably more input variables used (13/15 for the barrel/endcap versus the 8 for the trigger), the max depth was necessarily higher. Consequently, the learning rate was also made lower. The optimum hyperparameters were, then, found to be a maximum depth of 18 and a learning rate of 0.06.

A Data/MC BDT

While the above $\gamma + Jet$ BDT can help differentiate real photon candidates from fakes, it cannot, in principle, properly differentiate a photon candidate coming from a signal process from a photon candidate coming from a background process. Since the EGamma trigger selections will already cut out many events with jet-like photon candidates, much of the $H \rightarrow \gamma\gamma$ preselection is aimed at removing real photons coming from non-resonant processes (or even non-Higgs resonances). For the standard preselection, the p_T and $p_T/M_{\gamma\gamma}$ cuts achieve this.

Furthermore, the $\gamma + Jet$ BDT does not properly mirror the MVA-HLT; since the trigger-level BDT was training using Higgs MC as the signal and real data as the background, the offline selection should employ a similar strategy. So, in the same way that the trigger was trained using signal photon candidates originating from a ggH MC sample and background photon candidates originating from the data. The same input variables used in the $\gamma + Jet$ BDT were then used to train and optimize the new model.

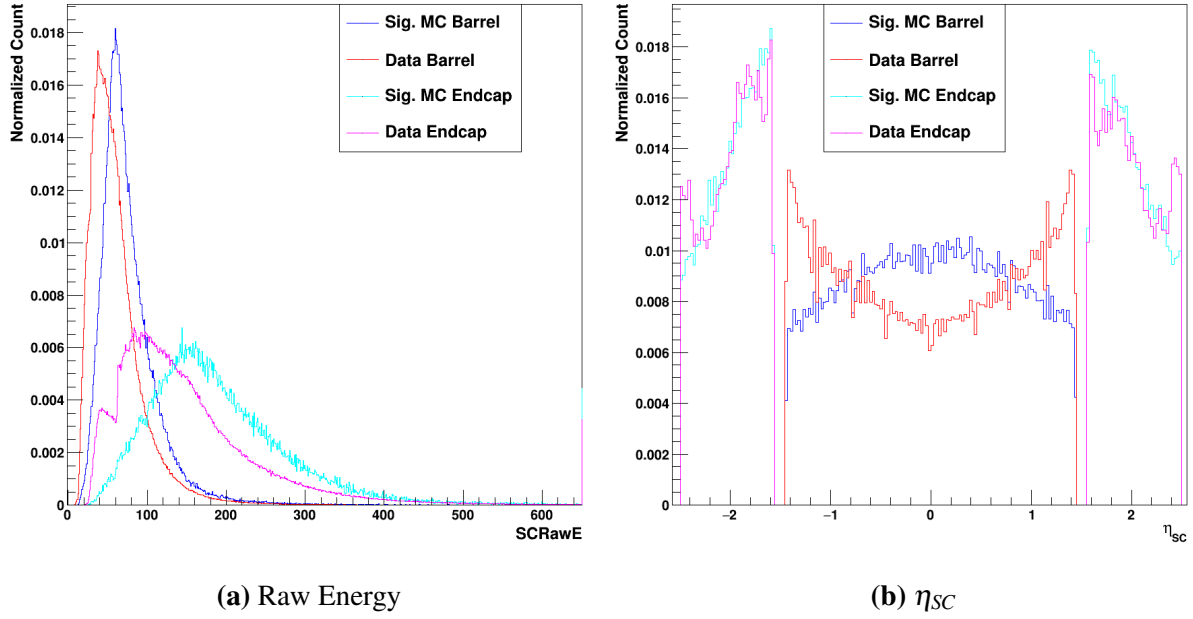


Figure 6.8. Plots of the Supercluster raw energy and η_{SC} . The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

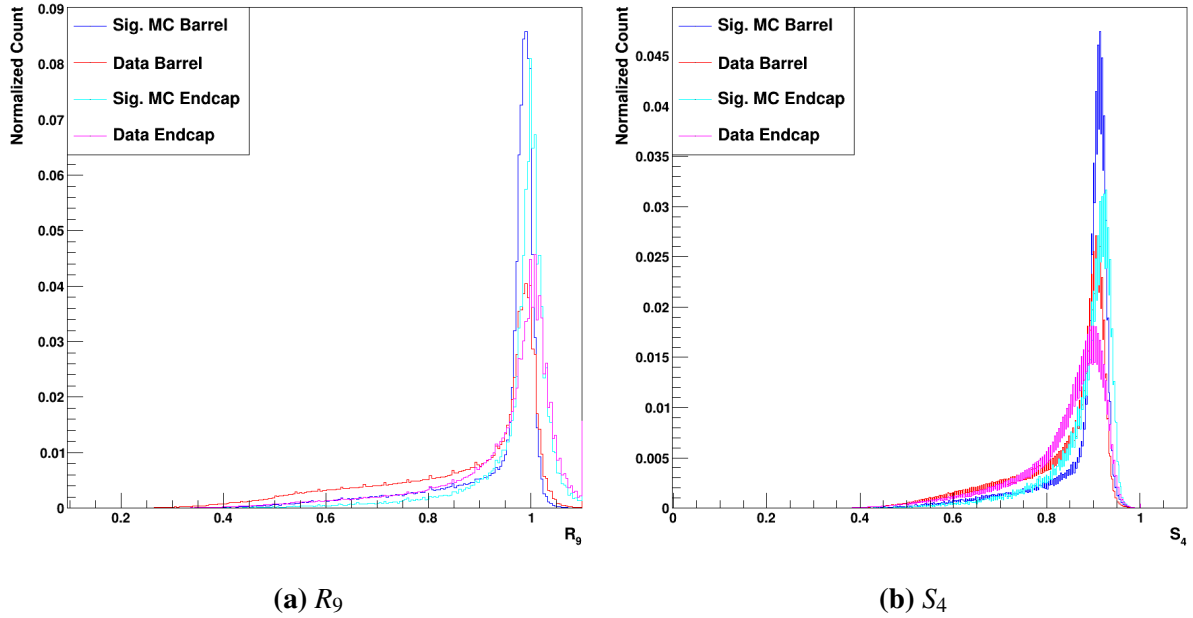


Figure 6.9. Plots of the R_9 and S_4 . The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

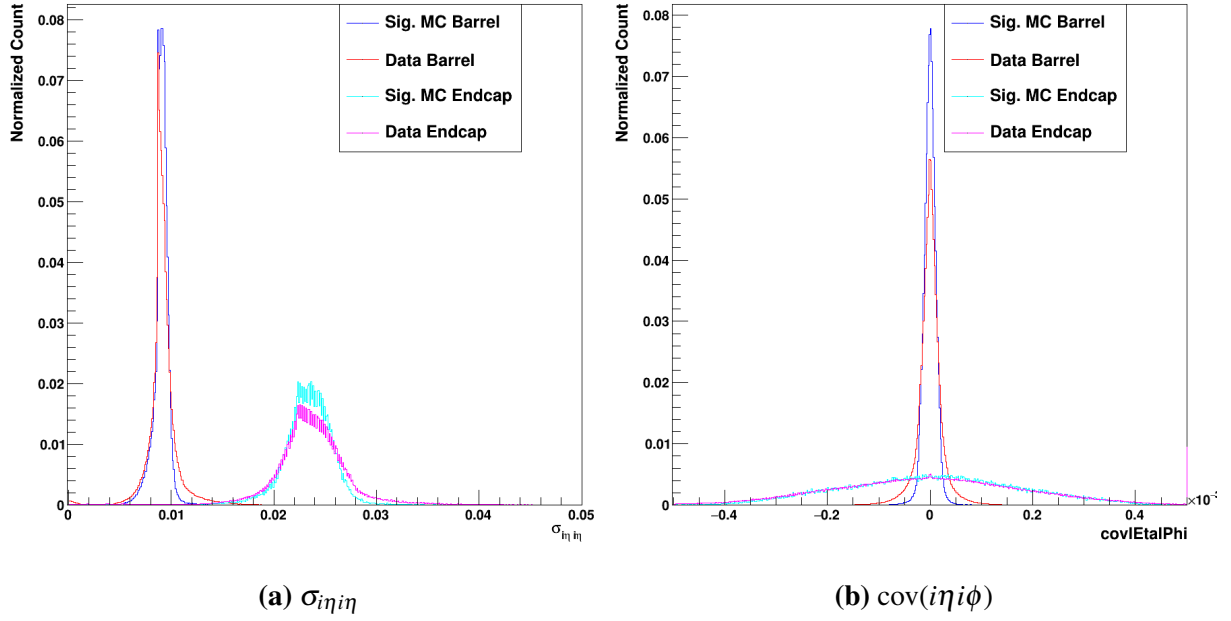


Figure 6.10. Plots of the $\sigma_{i\eta i\eta}$ and $\text{cov}(i\eta i\phi)$. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

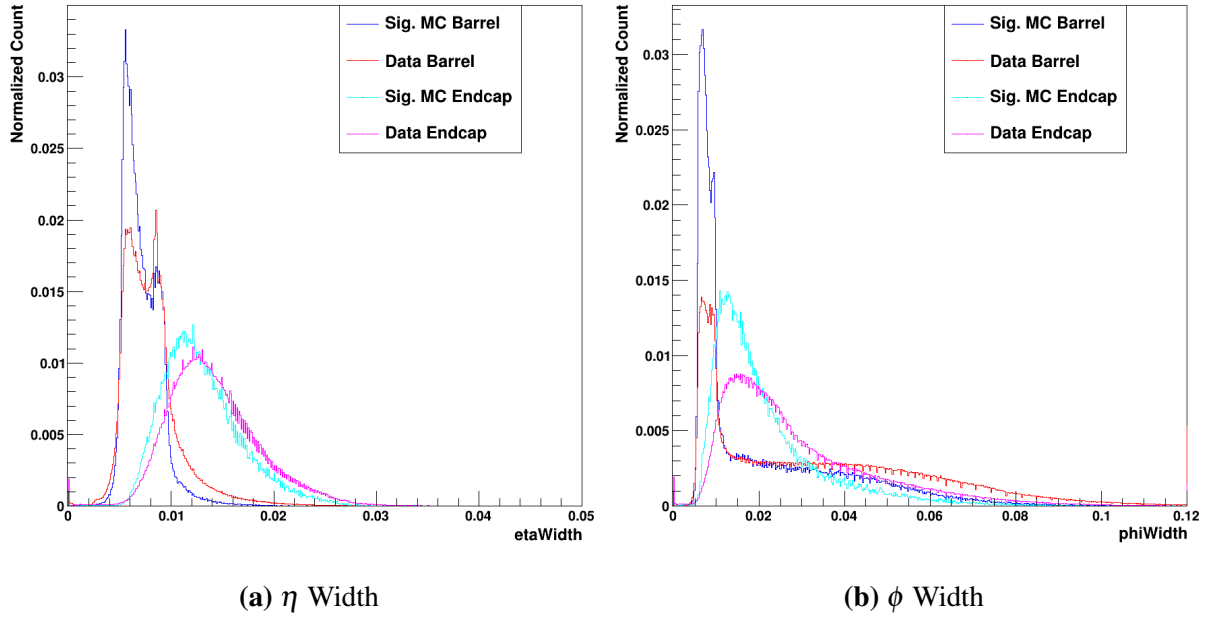
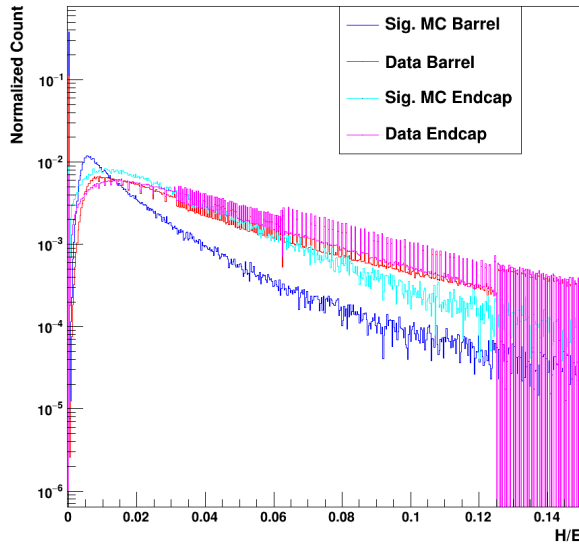
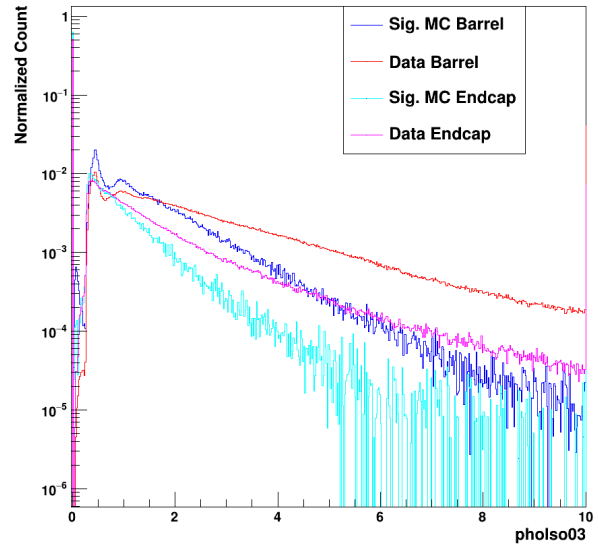


Figure 6.11. Plots of the η and ϕ widths. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

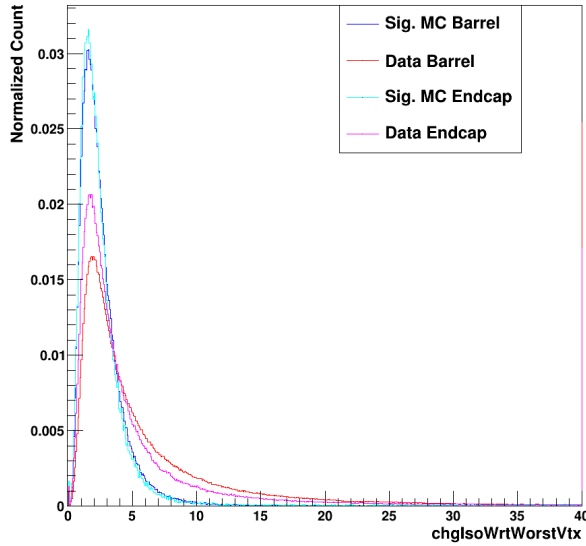


(a) H/E

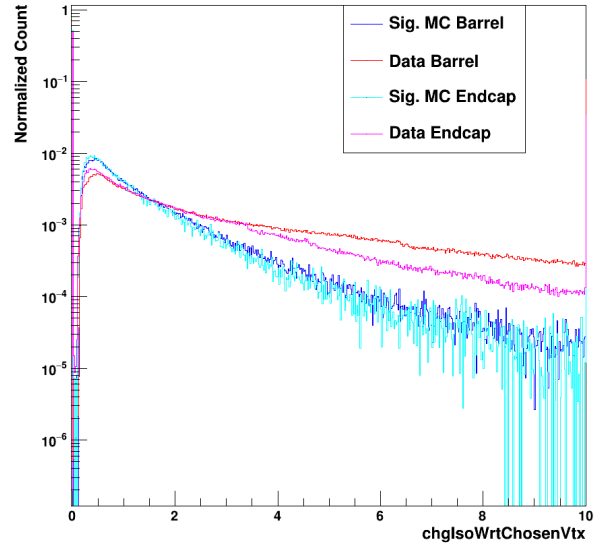


(b) Photon Isolation

Figure 6.12. Plots of the H/E and photon isolation. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.



(a) Chg. Iso. W.r.t. Worst Vertex



(b) Chg. Iso. W.r.t. Chosen Vertex

Figure 6.13. Plots of the charged isolations with respect to the chosen and worst vertices. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

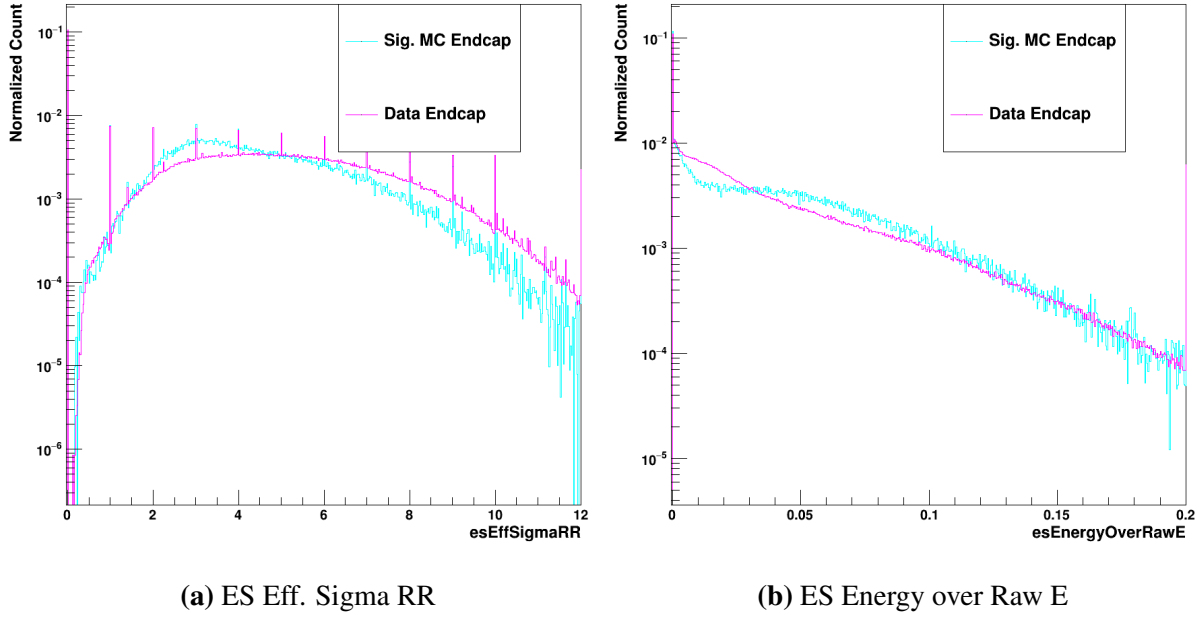


Figure 6.14. Plots of two endcap-specific variables: esEffSigmaRR and esEnergyOverRawE. The prompt signal is shown in light blue, whereas the fake background is shown in pink.

The same process of hyperparameter optimization as the previous models was used.

6.2.2 Determination of Loose Single Photon MVA Cuts

The preexisting preselection first makes decently loose cuts on individual photons, then tighter p_T and $p_T/M_{\gamma\gamma}$ cuts on selected diphoton combinations. A very similar strategy can be used for the BDT-based offline selection. Firstly, loose cuts on the above BDT scores can be applied to the individual photon candidates, which will reduce the number of possible diphoton combinations and cut out many events with no reasonable photon candidates. Then, with the list of individual photons cleaned up a bit, a tighter selection can be applied to the resulting diphoton objects for the selection of a final pair of photons.

So, then, the aim is to choose a loose set of cuts on the $\gamma + Jet$ and Data/MC BDT scores before applying a tighter final selection on the resulting diphoton combinations. One great advantage of a BDT-based approach is the ease with which these cuts can be adjusted and optimized; while a many-variable preselection has several different cuts to optimize, a binary

classifier BDT only has a single output score on which to make cuts. These cuts were, for now,

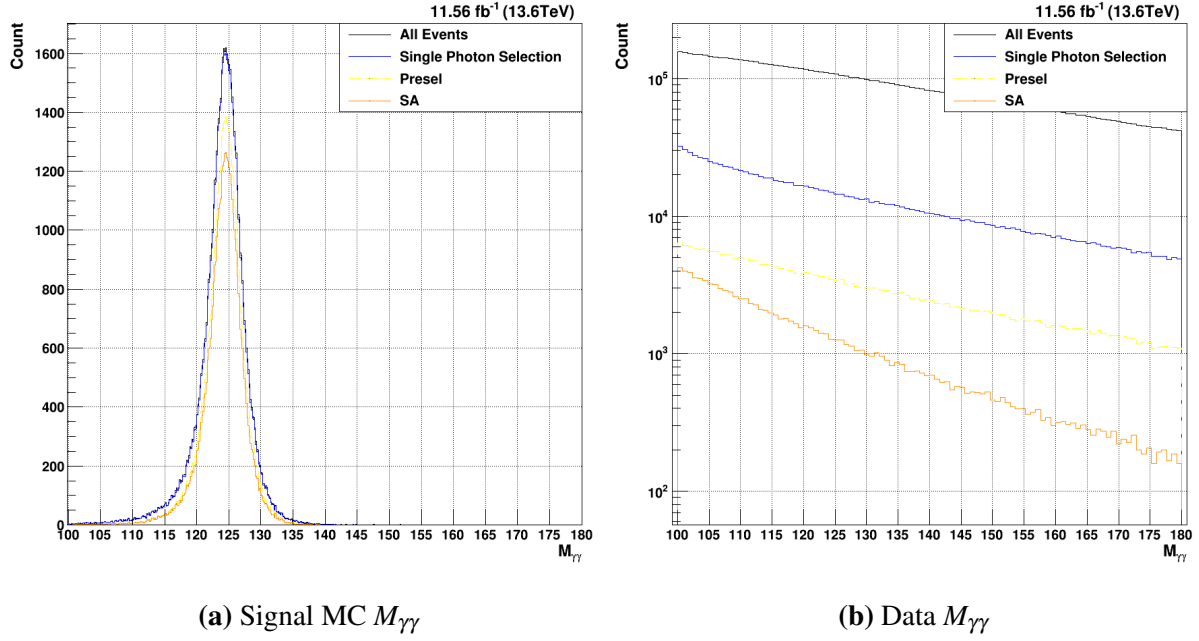


Figure 6.15. Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for $M_{\gamma\gamma}$

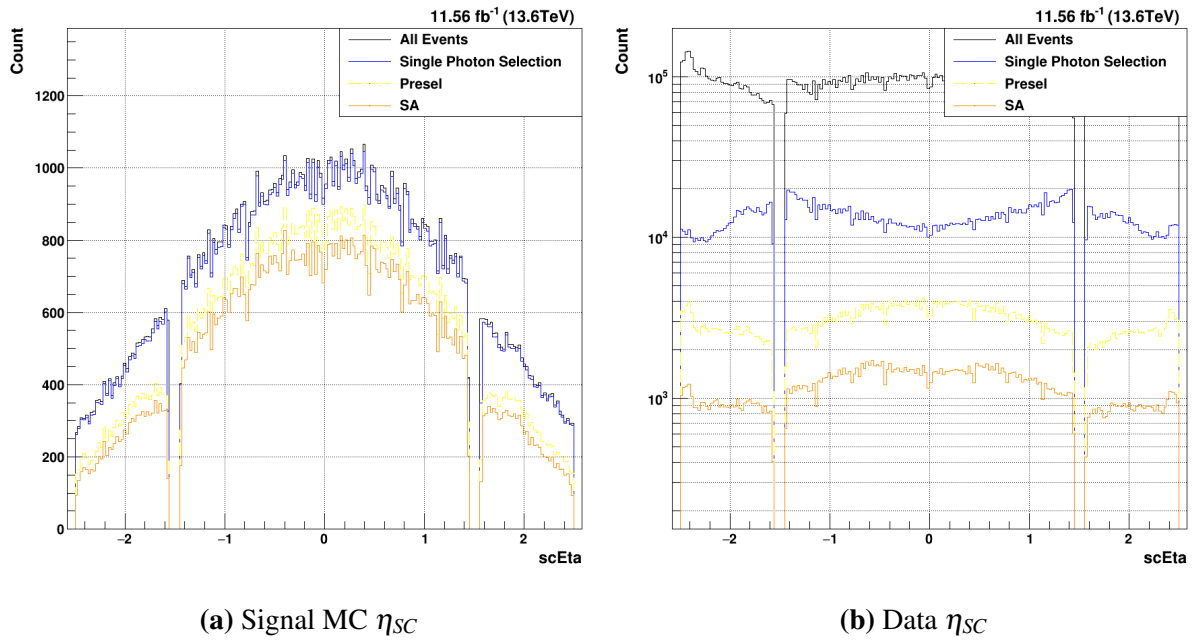


Figure 6.16. Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for η_{sc} . Both the lead and sublead photons are plotted together.

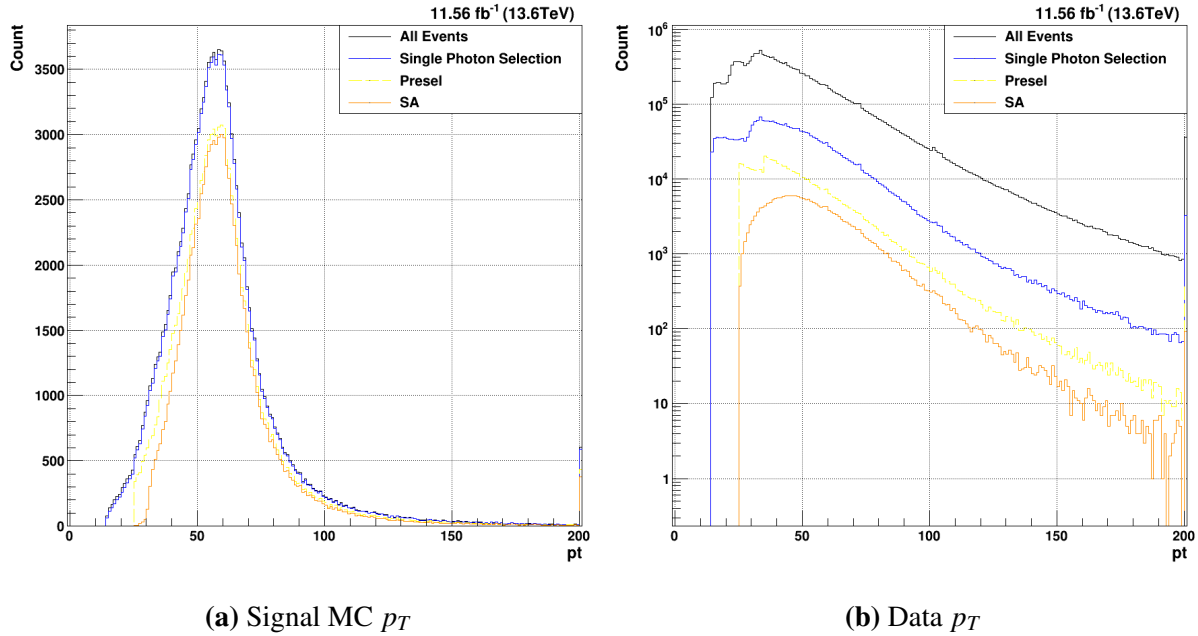


Figure 6.17. Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for p_T . Both the lead and sublead photons are plotted together.

6.2.3 Creation of Diphoton BDT

With the single photon preselection cuts determined, an additional selection must be placed on the diphoton objects. The most obvious continuation of the current strategy is to create an additional MVA that now works on diphoton quantities. Previous analysis strategies have, in fact, employed a diphoton MVA for the classification of events [12]. A new diphoton BDT can be constructed in reference to this, with the aim of making final cuts on this score such that the background rate is controlled while maintaining maximum signal efficiency.

While the goal of the single photon BDTs is to reject individual candidates that appear to not be photons, the diphoton BDT aims to properly differentiate diphoton candidates coming from $H \rightarrow \gamma\gamma$ from both events that do not contain two real photons and events with two real photons originating from background processes. As a result, the inputs to this model will consist primarily of lead and sublead single photon variables. By providing both the lead and sublead values to one model, correlations can be found that ensure that selected events contain two real photons with the necessary kinematics to possibly comprise a signal event. Below, in table 6.2,

Table 6.2. Table of diphoton BDT input variable names, and a short description of each.

Variable Name	Description
$p_T^{Lead}/M_{\gamma\gamma}$ and $p_T^{Sub}/M_{\gamma\gamma}$	The mass-normalized p_T for each photon
η_{sc}^{Lead} and η_{sc}^{Sub}	The supercluster η for each photon
$BDT_{\gamma+jet}^{Lead}$ and $BDT_{\gamma+jet}^{Sub}$	The $\gamma + jet$ BDT score for each photon
$BDT_{Data/MC}^{Lead}$ and $BDT_{Data/MC}^{Sub}$	The Data/MC BDT score for each photon
Lead and Sublead Electron Veto	A boolean to reject photons originating from electrons
$\cos(\Delta\phi)$	The cosine of the ϕ separation of the two photons
$\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$	A relative mass resolution estimate

each of the input variables is listed and described. These were largely adopted from the Run2 Diphoton photonID, with a few differences.

Since the goal of this model is to differentiate signal diphoton combinations from those in the background, signal MC and data can again be used for the signal and background for the BDT, as with the data/MC BDT described above. Again, the input variable distributions can be examined to see the separation between signal and background class events.

The first two variables are related directly to the kinematics of the diphoton object, and therefore are not split into lead and sublead values. The first, $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$, provides a relative invariant mass resolution estimate of the diphoton combination. It can clearly be seen in figure 6.18a, that there is great separation between the signal MC and data events. The signal MC tends to have a much lower $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$, which corresponds to a higher mass resolution. This is to be expected, as the two selected photons in the $H \rightarrow \gamma\gamma$ are much more likely to come from a resonant process than those in the data. The $\cos(\Delta\phi)$, however, does not exhibit the same strong separation. This variable aims to measure the separation in ϕ of the two selected photon candidates. Based on the other selections at play, however, it seems that this variable will not be a strong discrimination variable.

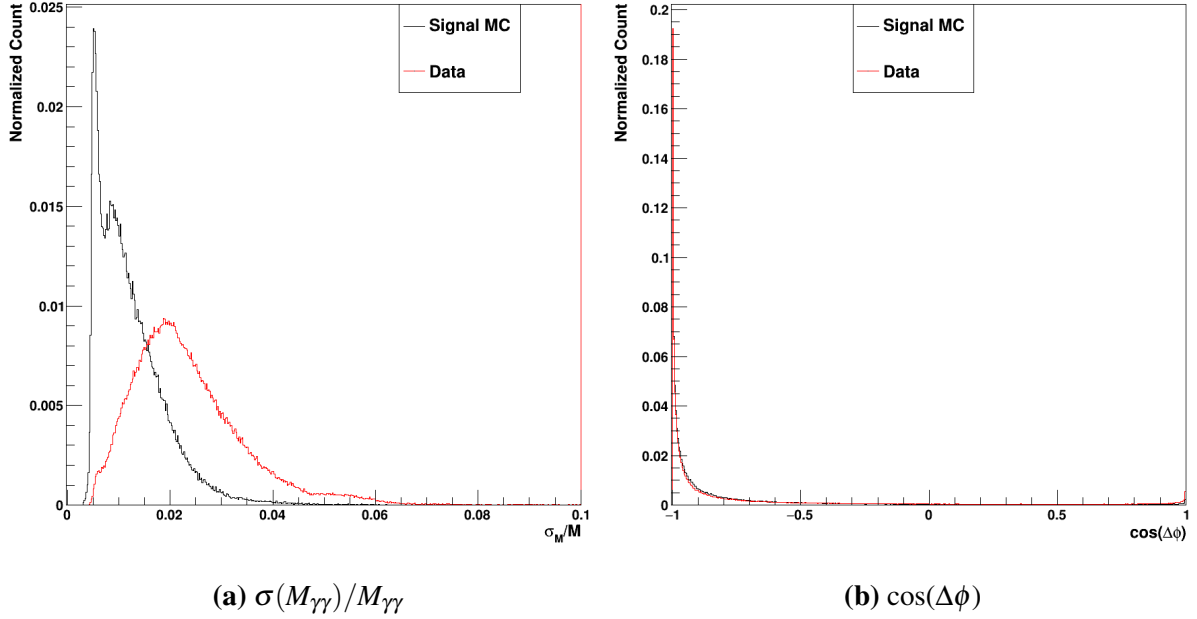


Figure 6.18. The two per-event diphoton input variables, $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ and $\cos(\Delta\phi)$. The signal MC is in black, whereas the data events are in red.

The next two pairs of input variables are the single photon BDT scores. Firstly, the γ + Jet single photon BDT scores are shown below in figure 6.19. As with all XGBoost BDT binary classification scores, a value near 0 shows that the algorithm has identified the photon as being part of the background class, while a score near 1 corresponds to an event identified as part of the signal class. For the signal MC, it can be seen that the scores peak very well near 1. This is to be expected, as the signal MC photons here are matched to a generated photon and therefore guaranteed to be a real photon. This effect is clear for both the lead and sublead photons.

Clearly, however, the background photons in red are not constrained entirely near 0. In fact, there are two peaks in both the lead and sublead score distributions. While many of the background photon candidates can easily be assigned a score near 0, there are also a large number for which the γ + Jet scores are not adequate to differentiate the background from the signal. This is likely due, at least in part, to the composition of the data overall. While many of the data events consist of dijet QCD background, there are likely many events for which one or even two real photons have been selected. This is exacerbated by the triggers required for this training

sample; here, since either the MVA-HLT trigger or the standard diphoton trigger must be passed to enter the training sample, the odds of a data event containing one or two real photons is much higher than it would be for other data samples. This demonstrate why the γ + Jet BDT alone is not adequate to correctly differentiate between signal and background events.

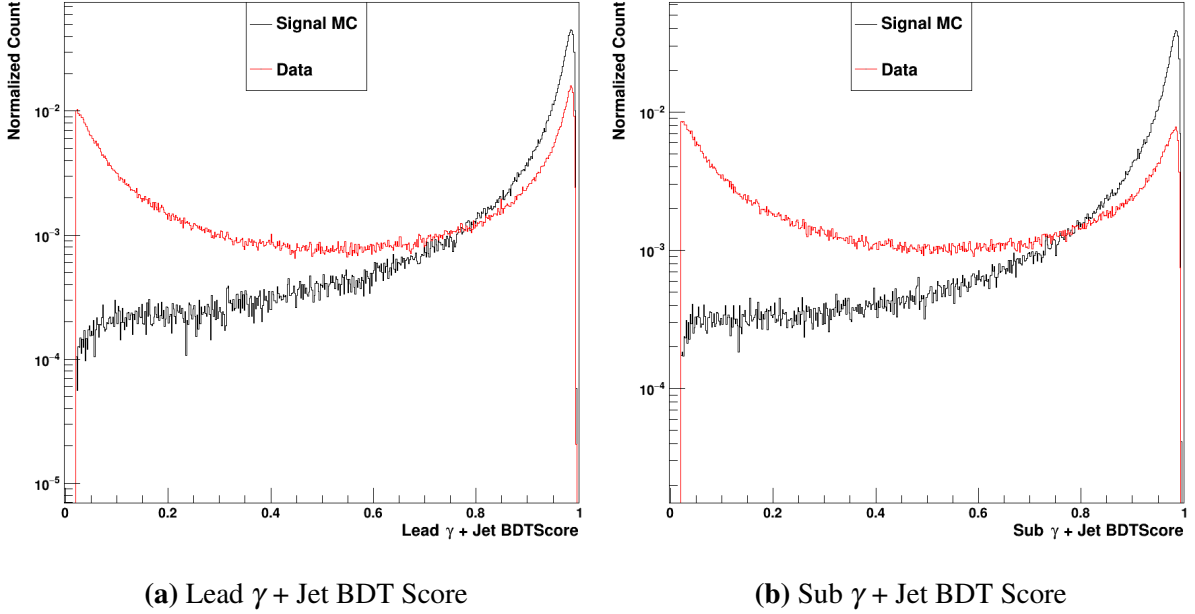


Figure 6.19. The lead and sublead γ + Jet XGBoost BDT scores. The signal MC is in black, whereas the data events are in red.

Secondly, the data/MC scores for both the lead and sublead photons are shown in figure 6.20 on a logarithmic scale. As expected, there is clear separation between the signal MC in black and the data in red. It is expected that the signal MC distributions will largely cluster near 1, which corresponds to a positive identification as signal. While this is true for both the lead and sublead scores, the identification of the background is perhaps a stronger effect. It can be seen that the data events are very closely clustered near 0. Very few of the background events, on the other hand, are placed near a score of 1.

Similarly to the γ + Jet single photon BDT, then, it is clear that this data/MC BDT is not adequate alone to differentiate signal and background. The diphoton BDT, then, is a crucial step of the signal and background selection; by having access to both the single photon BDTs,

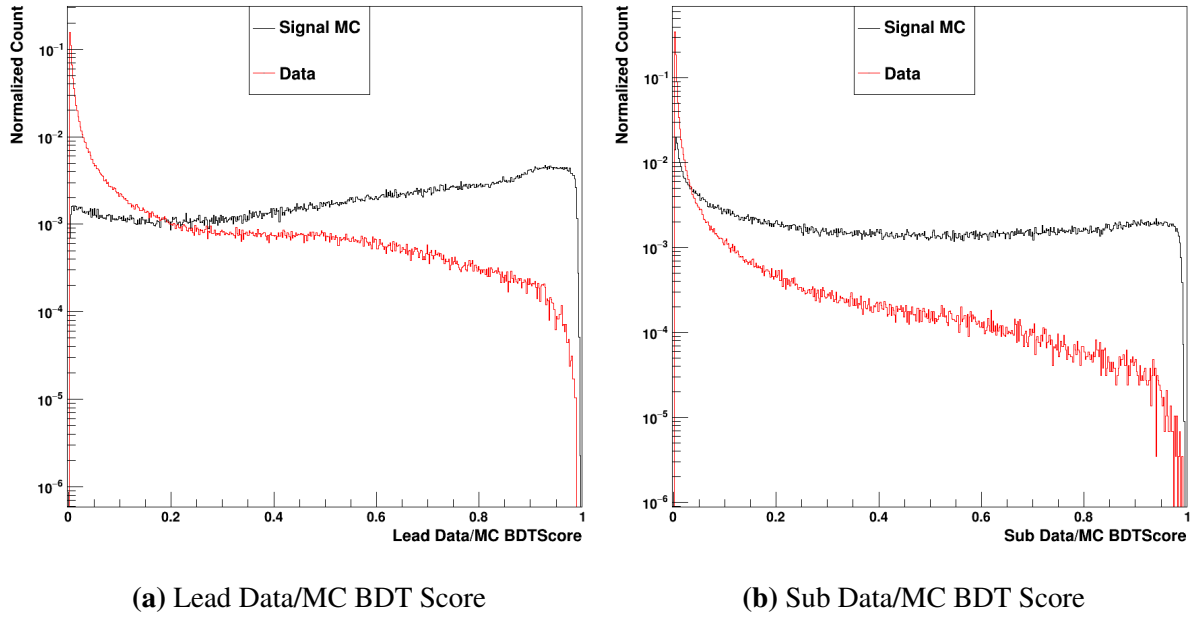
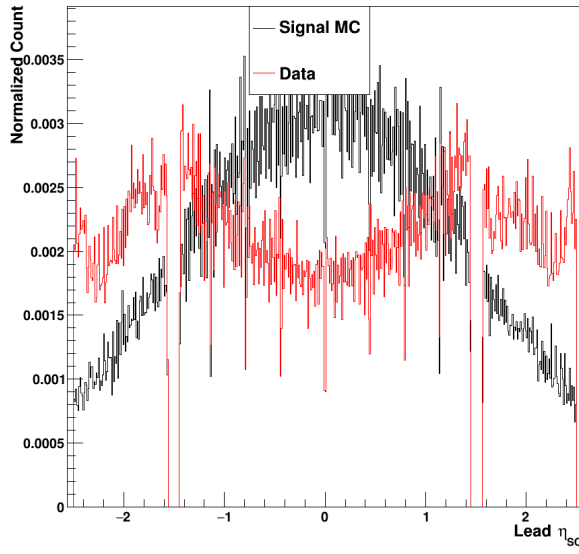


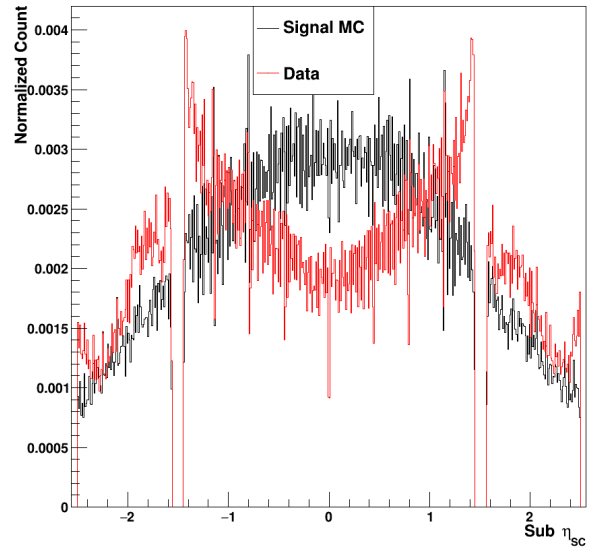
Figure 6.20. The lead and sublead Data/MC XGBoost BDT scores. The signal MC is in black, whereas the data events are in red.

the diphoton BDT can construct correlations that allow the final model to be considerably more discerning than input scores it uses. As will be shown in (REF?), this works to great effect.

While the single photon quality scores given by the single photon BDTs are quite helpful, the diphoton kinematics are additionall a very strong source of discrimination power.

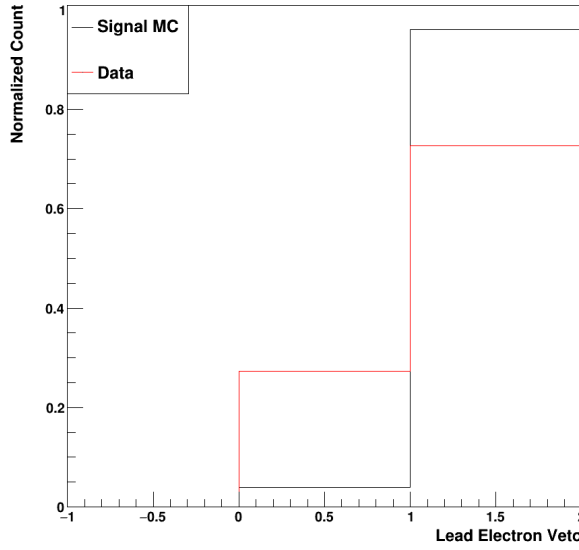


(a) Lead η_{SC}

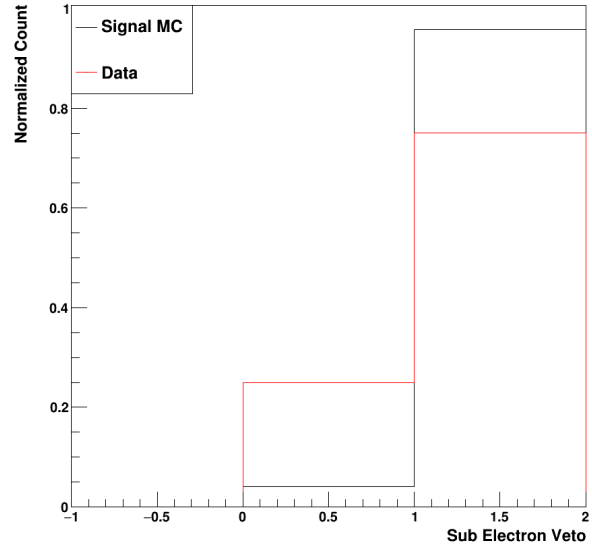


(b) Sub η_{SC}

Figure 6.21. The lead and sublead η_{SC} . The signal MC is in black, whereas the data events are in red.



(a) Lead Electron Veto



(b) Sub Electron Veto

Figure 6.22. The lead and sublead electron veto. The signal MC is in black, whereas the data events are in red.

Chapter 7

Evaluation of the MVA-HLT and Complementary Offline Selection

7.1 Measurement of Trigger Rate

As discussed in chapter 5, the first main consideration for any HLT is the trigger rate, usually measured in Hz. There are two main corresponding numbers: the total rate measures the total number of events being accepted by the trigger, whereas the pure rate measures the number of events being accepted by a trigger that would not be accepted by any other trigger.

7.1.1 Method of Rate Measurement

7.1.2 Trigger Rate of MVA-HLT

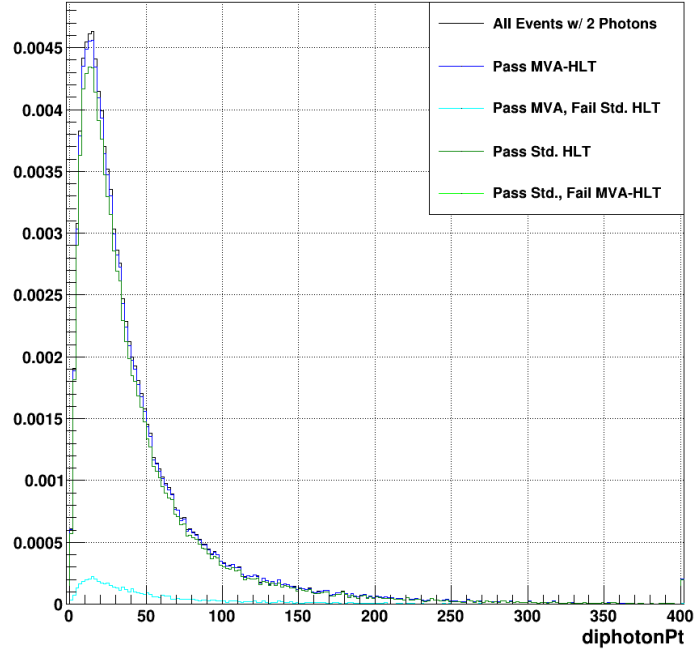
7.2 Evaluation of MVA-HLT Efficiency on Selected Signal Samples

Once it has been verified that the trigger rate is within the limits posed by the collaboration, the next major task is ensuring that the signal efficiency for the MVA-HLT is a noticeable improvement over the standard HLT. Since the reconstruction, calibrations, and corrections are different at the NanoAOD level, the measurements made at the RAW level in section 5.3 must be verified on independent analysis-level (NanoAOD) samples.

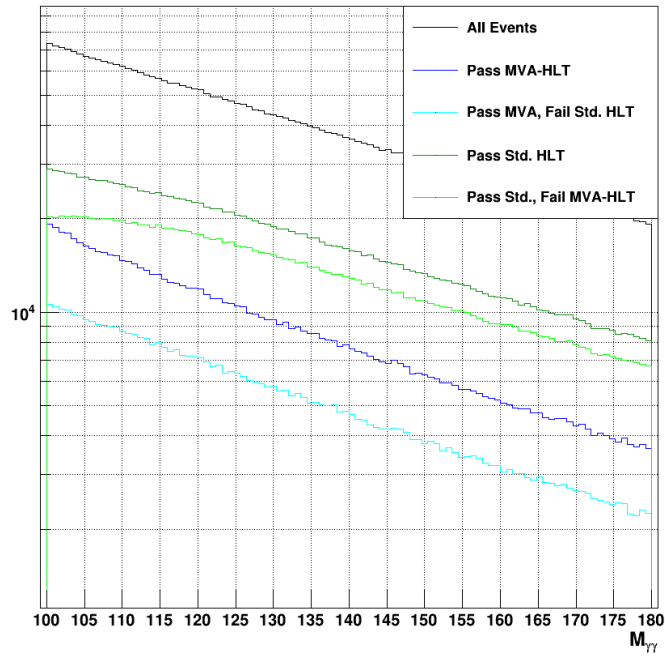
As in section 5.3, plots can be made of selected input variables and the resulting accepted

distributions can be compared for the MVA-HLT versus the standard diphoton HLT.

7.2.1 Single Higgs Signal Efficiency

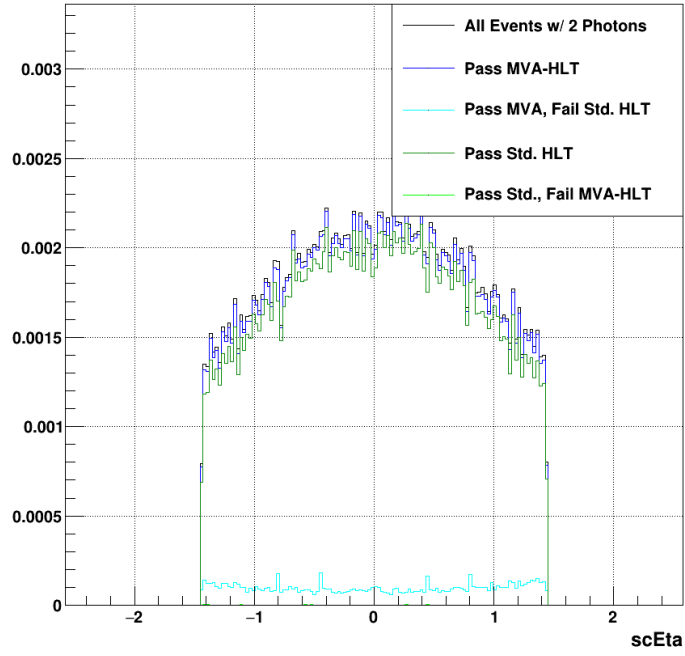


(a)

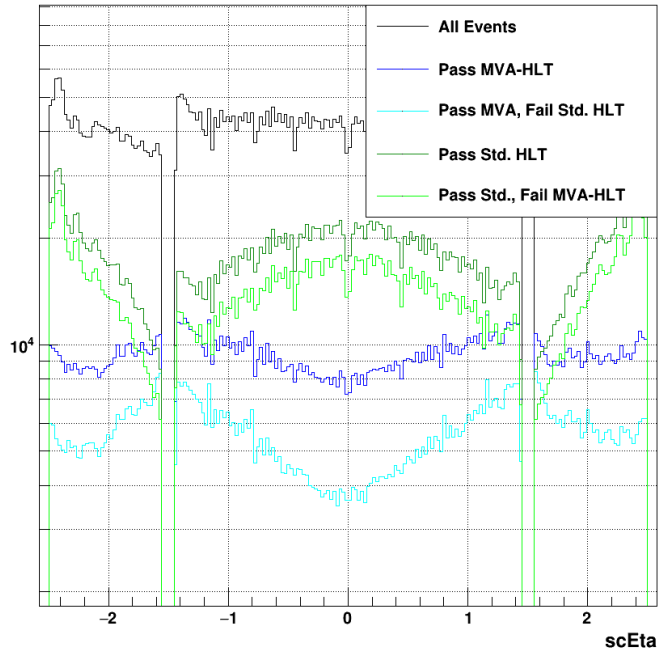


(b)

Figure 7.1. Diphoton invariant mass for the signal MC (left) and data (right). Events accepted by the MVA-HLT are shown in dark blue, while events accepted by the standard HLT are shown in dark green.

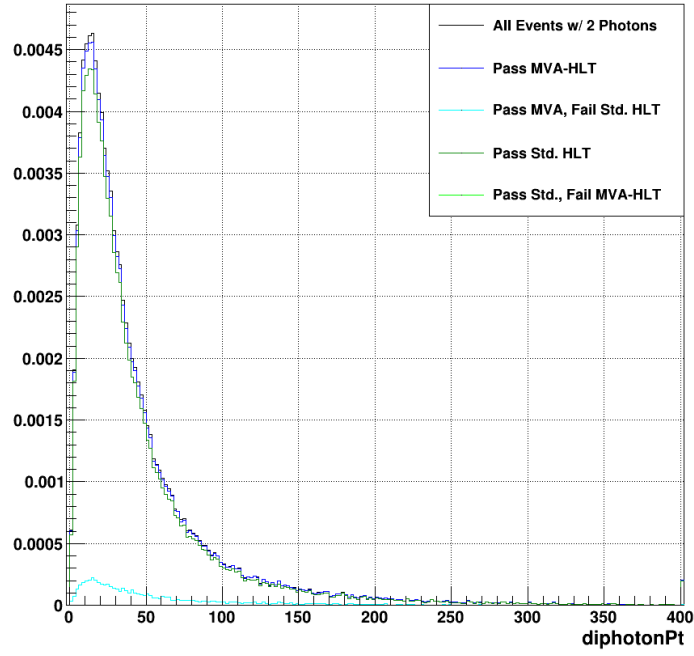


(a)

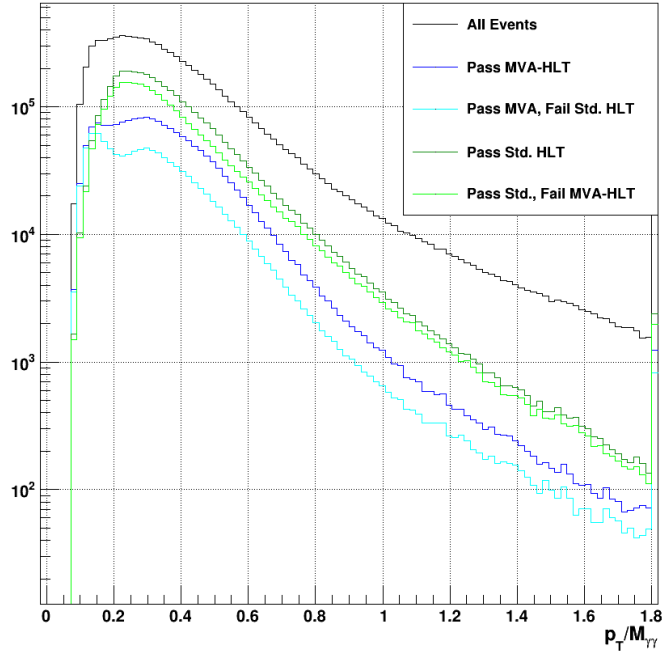


(b)

Figure 7.2. The η_{sc} of both photons for the signal MC (left) and data (right). Events accepted by the MVA-HLT are shown in dark blue, while events accepted by the standard HLT are shown in dark green.



(a)



(b)

Figure 7.3. The $p_T/M_{\gamma\gamma}$ of both photons for the signal MC (left) and data (right). Events accepted by the MVA-HLT are shown in dark blue, while events accepted by the standard HLT are shown in dark green.

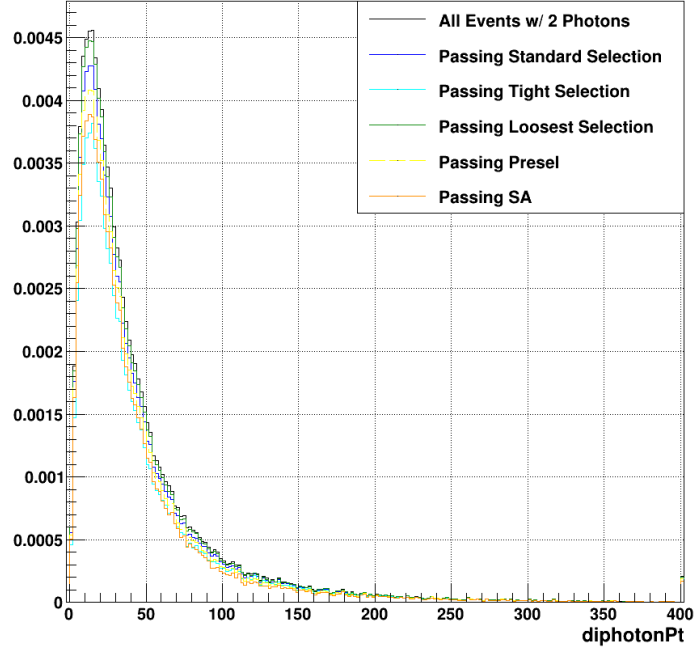
7.2.2 DiHiggs Signal Efficiency

7.3 Evaluation of Offline Selection Efficiency with MVA-HLT

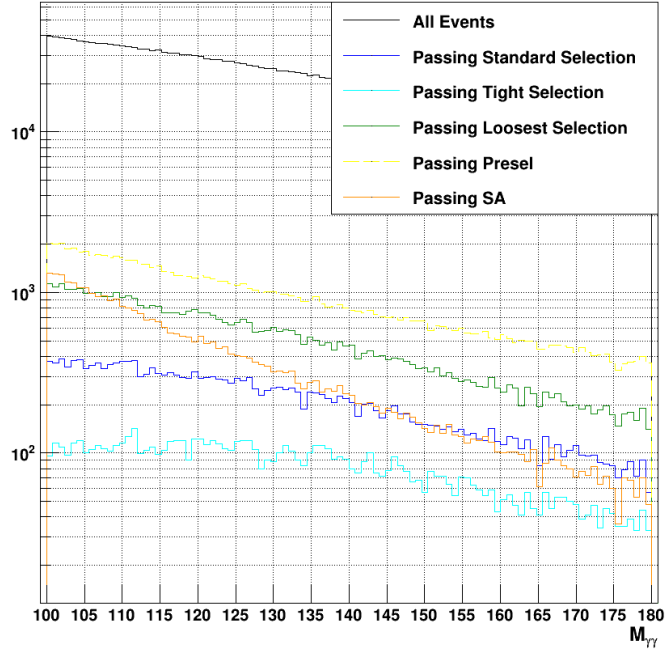
As discussed in chapter 6, no HLT exists

7.3.1 Method of Signal Efficiency Measurement

7.3.2 Single Higgs Signal Efficiency

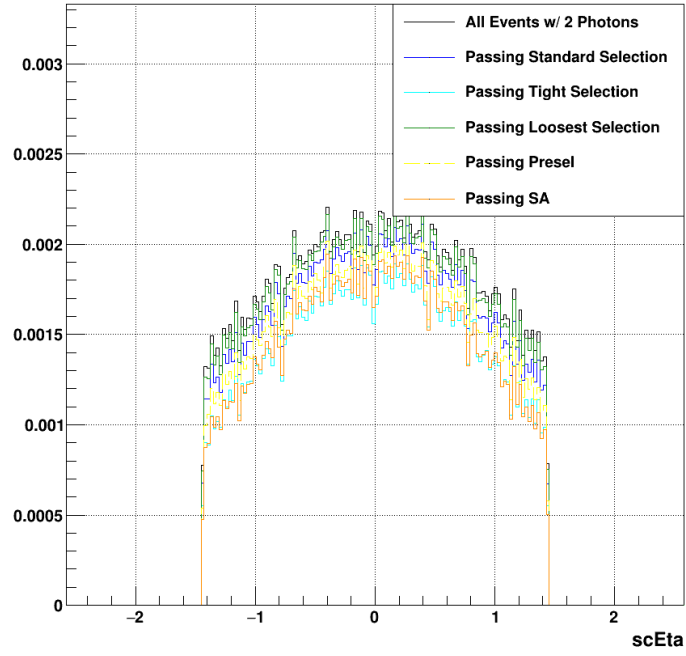


(a)

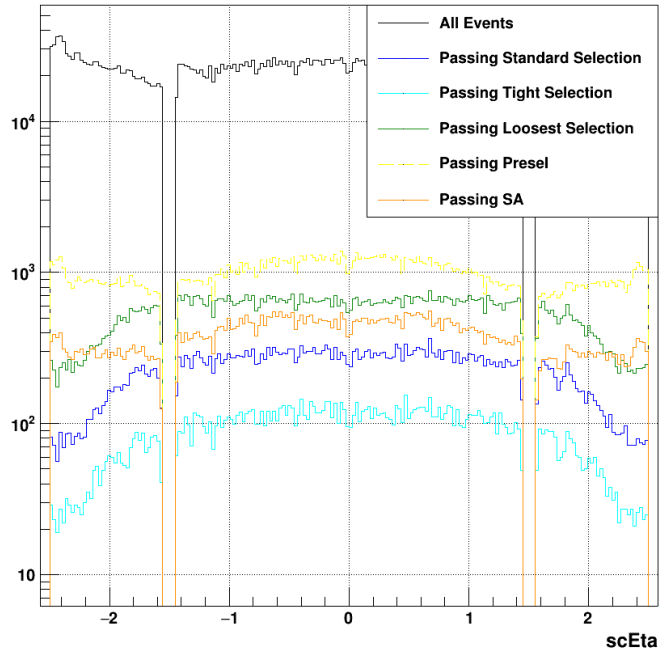


(b)

Figure 7.4. Diphoton invariant mass for the signal MC (left) and data (right). Events accepted by the MVA-HLT are shown in dark blue, while events accepted by the standard HLT are shown in dark green.

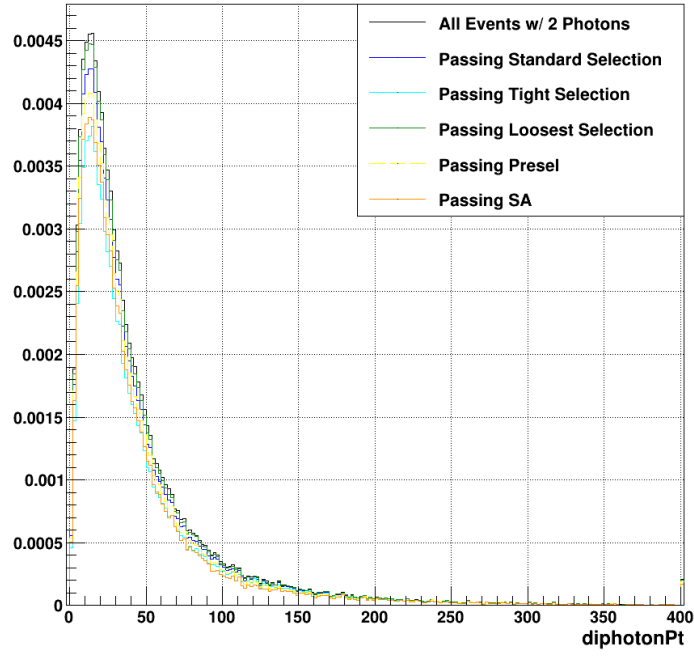


(a)

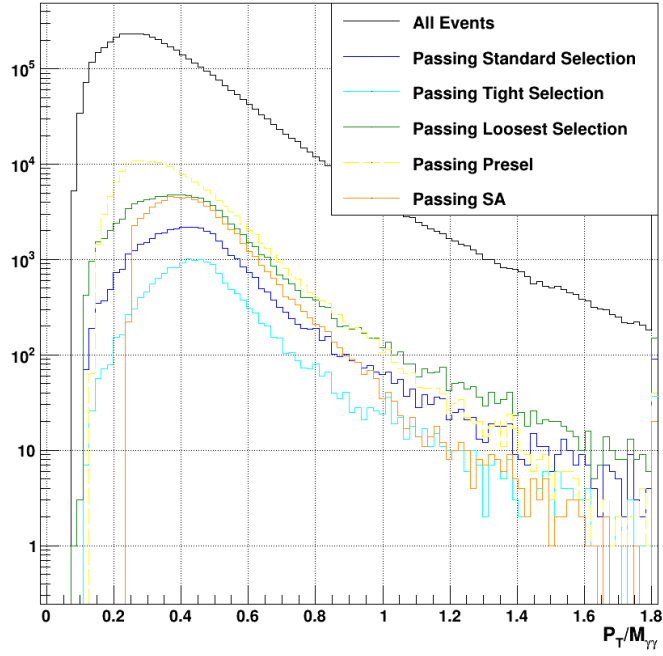


(b)

Figure 7.5. The η_{sc} of both photons for the signal MC (left) and data (right). Events accepted by the MVA-HLT are shown in dark blue, while events accepted by the standard HLT are shown in dark green.



(a)



(b)

Figure 7.6. The $p_T/M_{\gamma\gamma}$ of both photons for the signal MC (left) and data (right). Events accepted by the MVA-HLT are shown in dark blue, while events accepted by the standard HLT are shown in dark green.

7.3.3 DiHiggs Signal Efficiency

Chapter 8

Summary

Chapter 9

Acknowledgements

Appendix

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